

Sep. 11, 2025

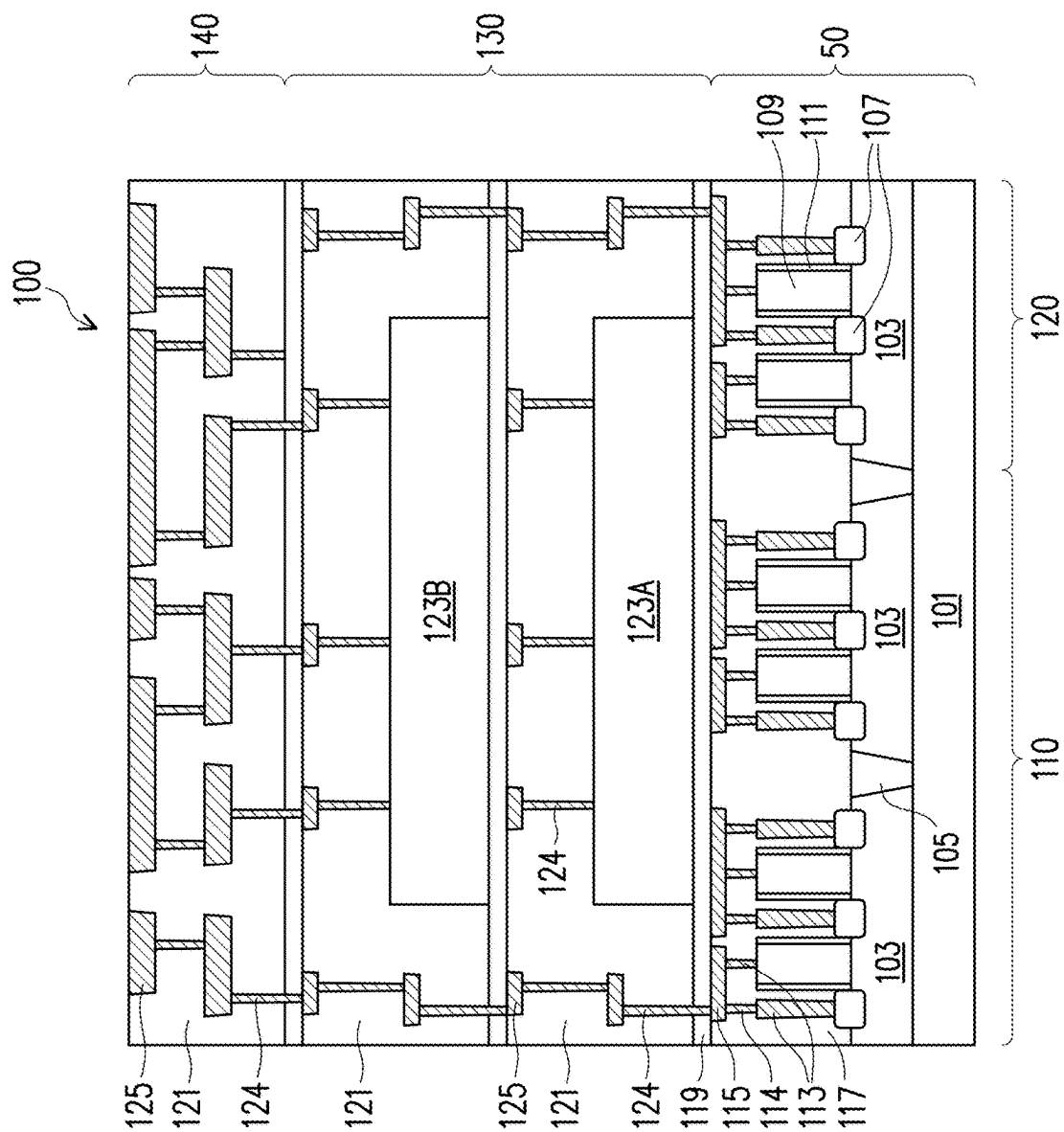


FIG. 1

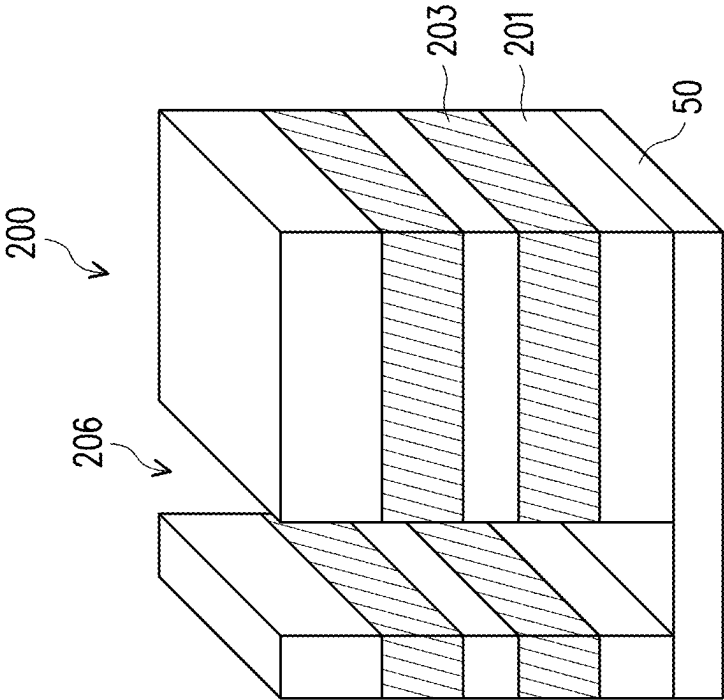


FIG. 2

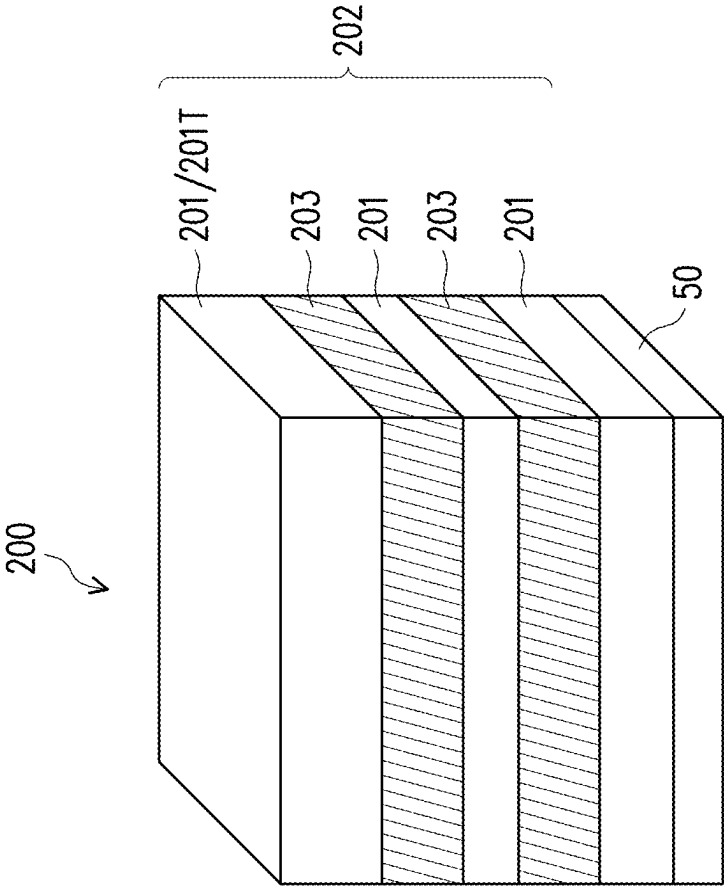


FIG. 3

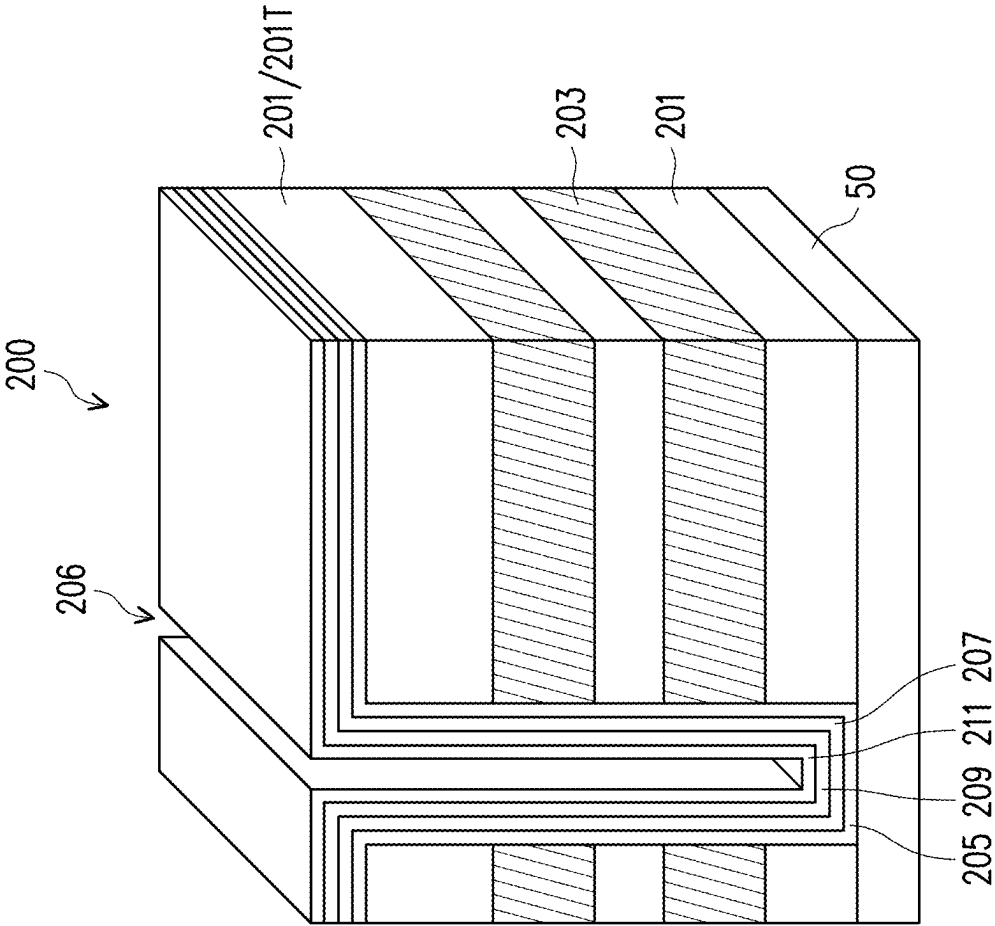


FIG. 4

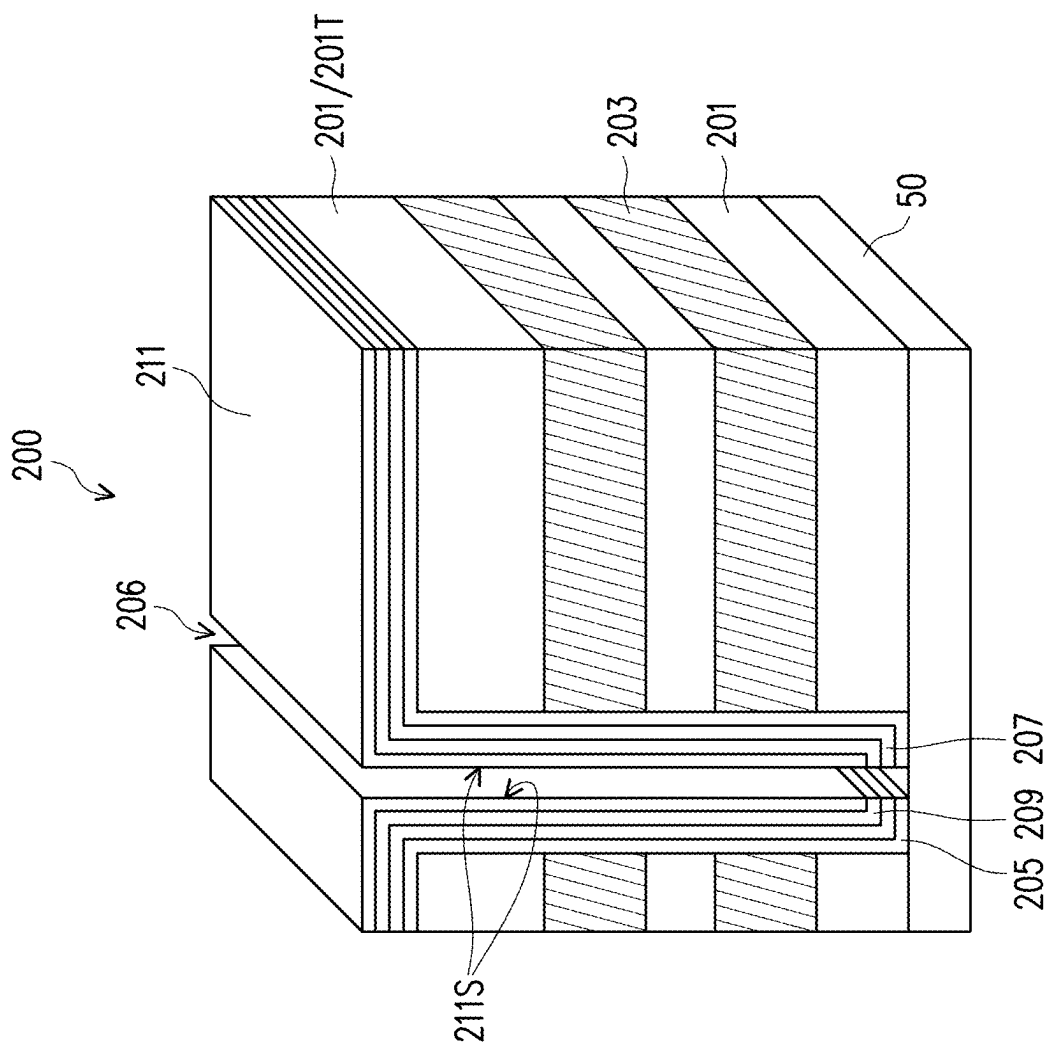


FIG. 5

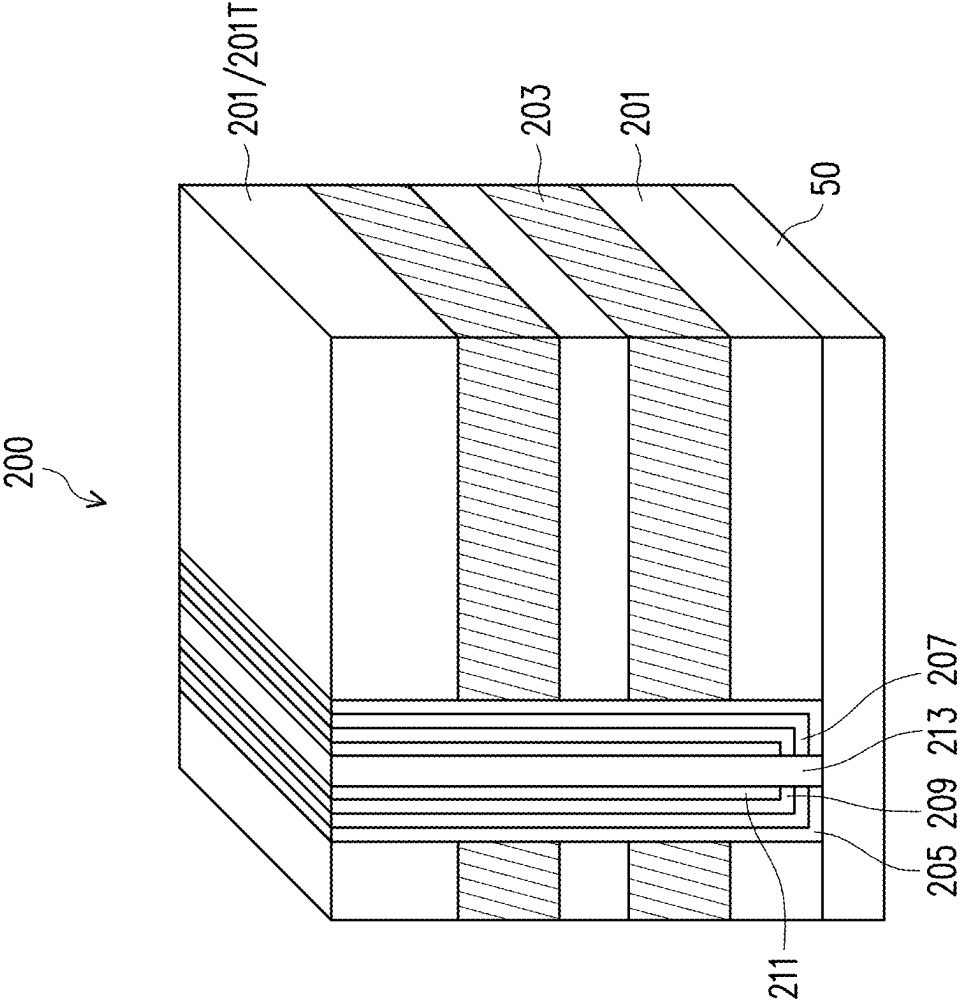


FIG. 6

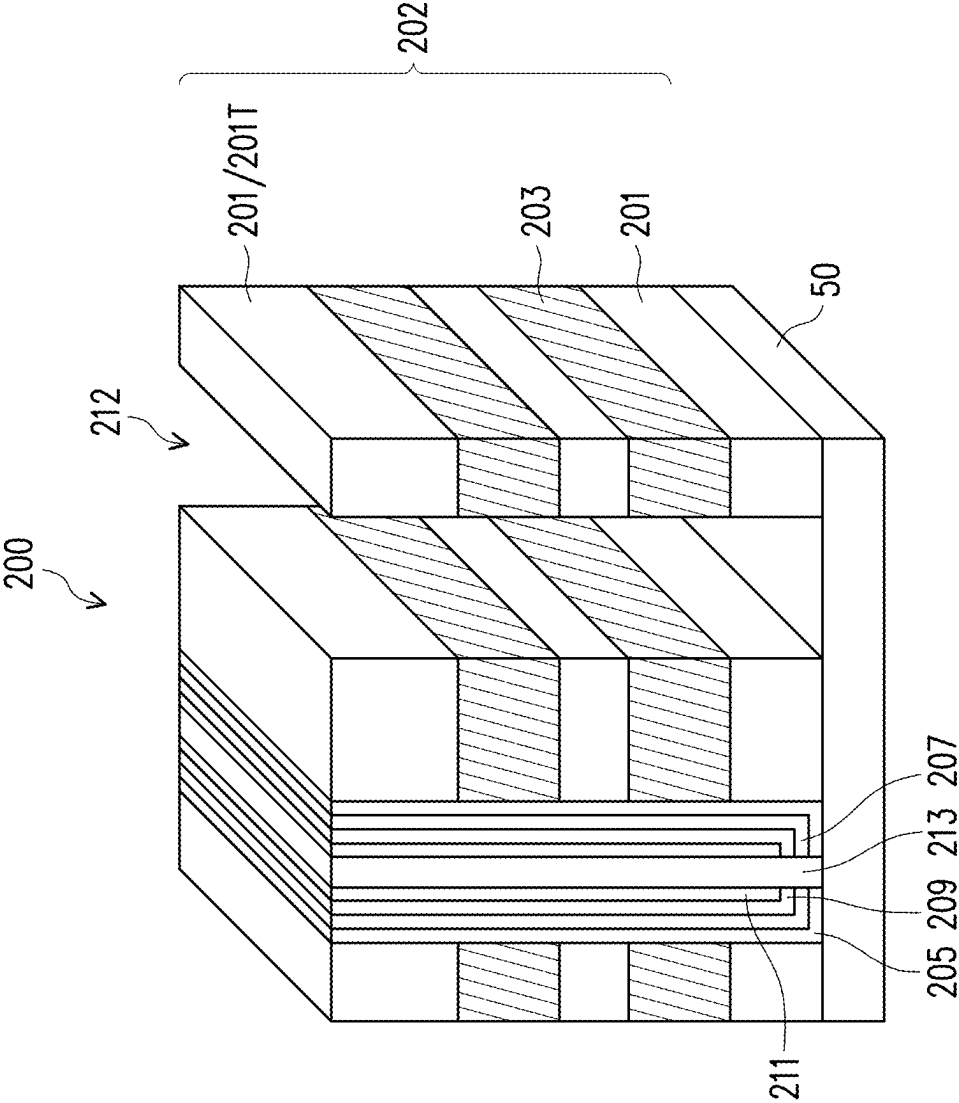


FIG. 7

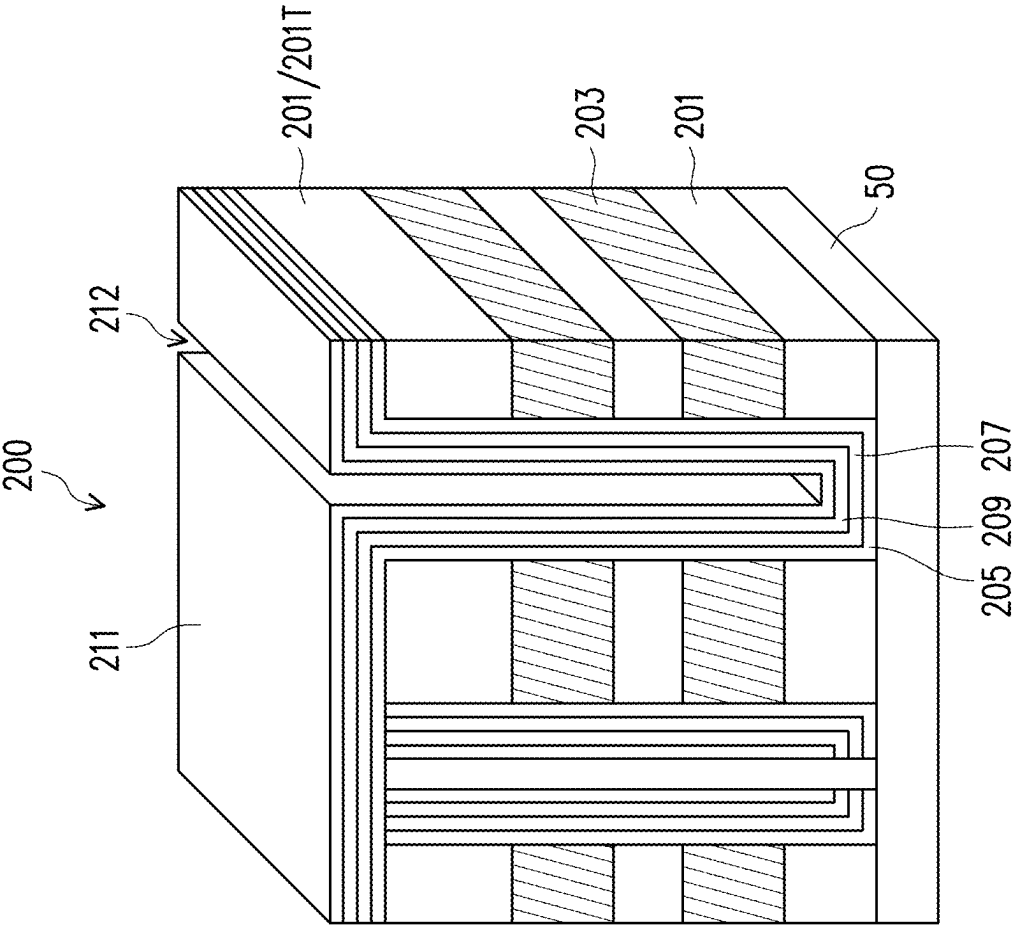


FIG. 8

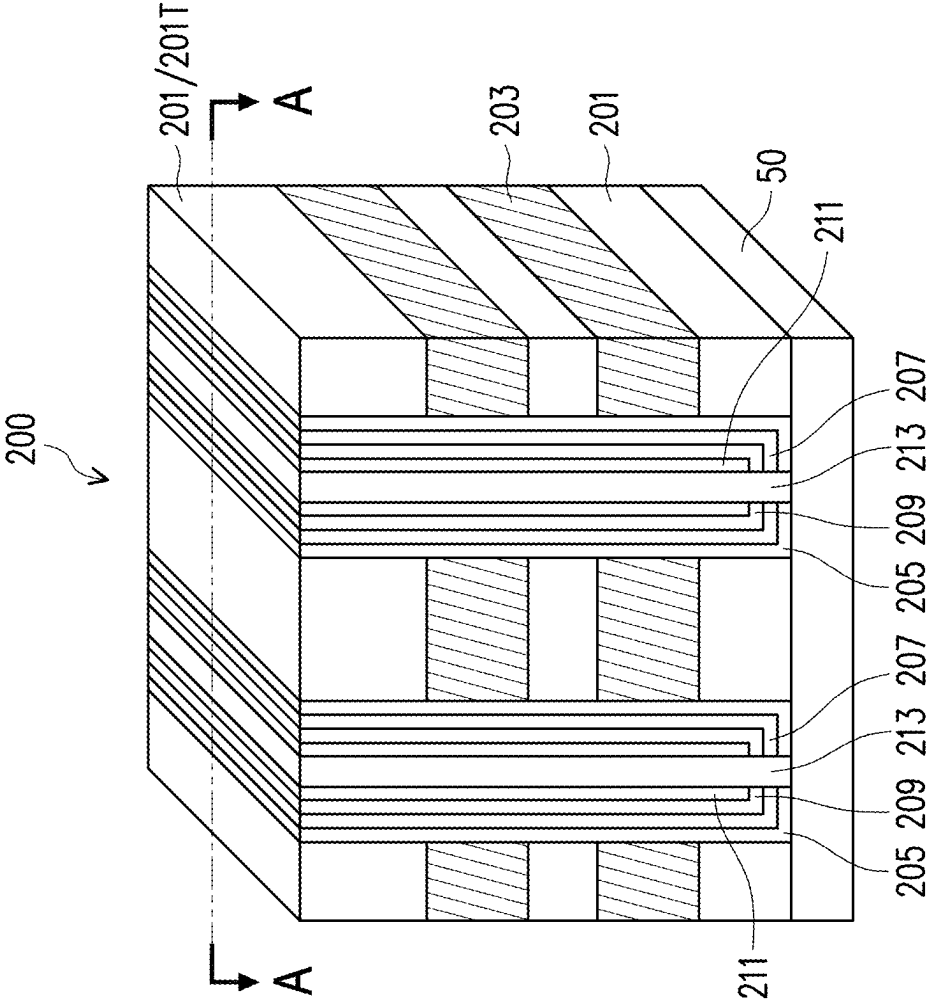


FIG. 9A

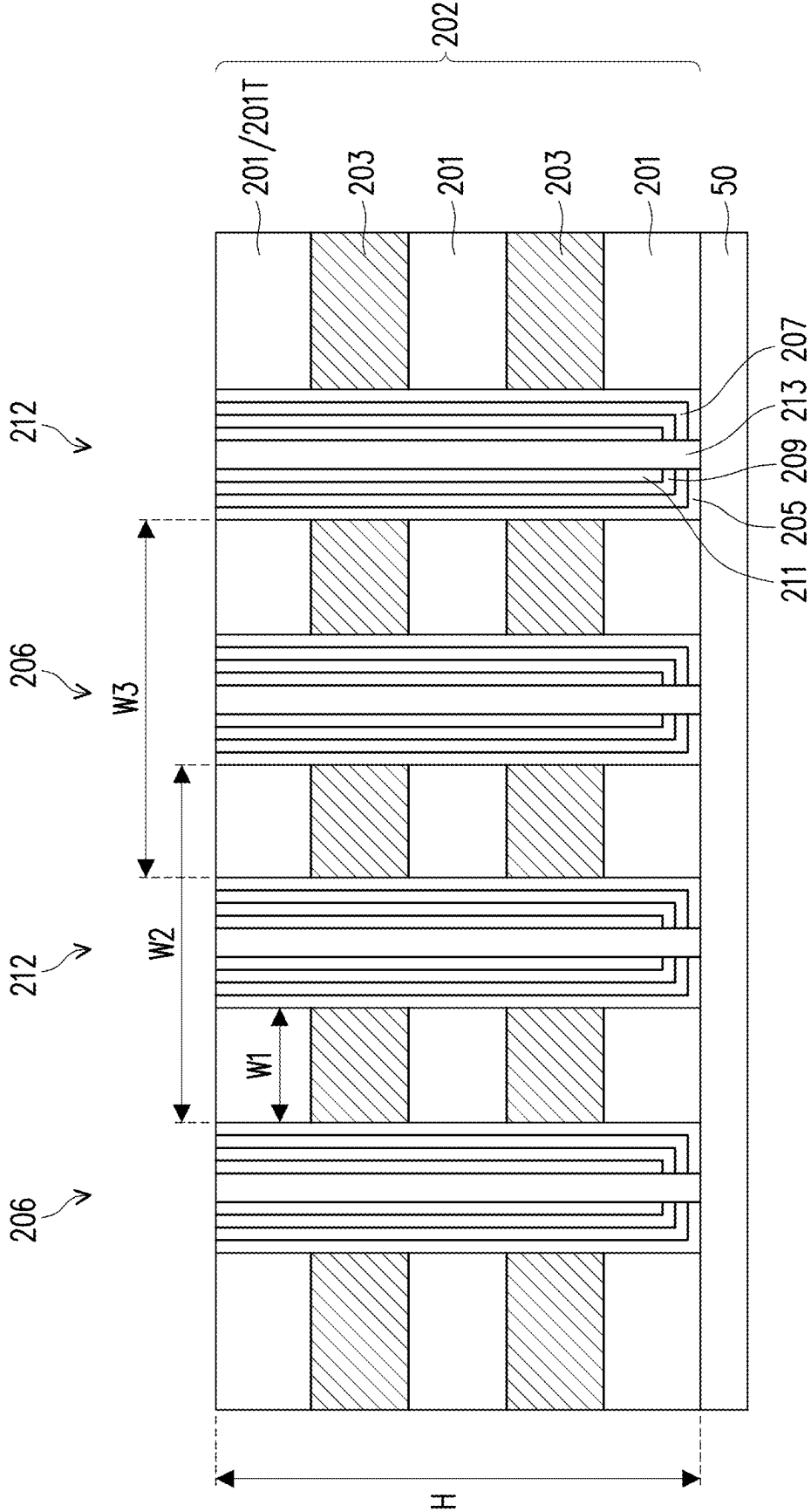


FIG. 9B

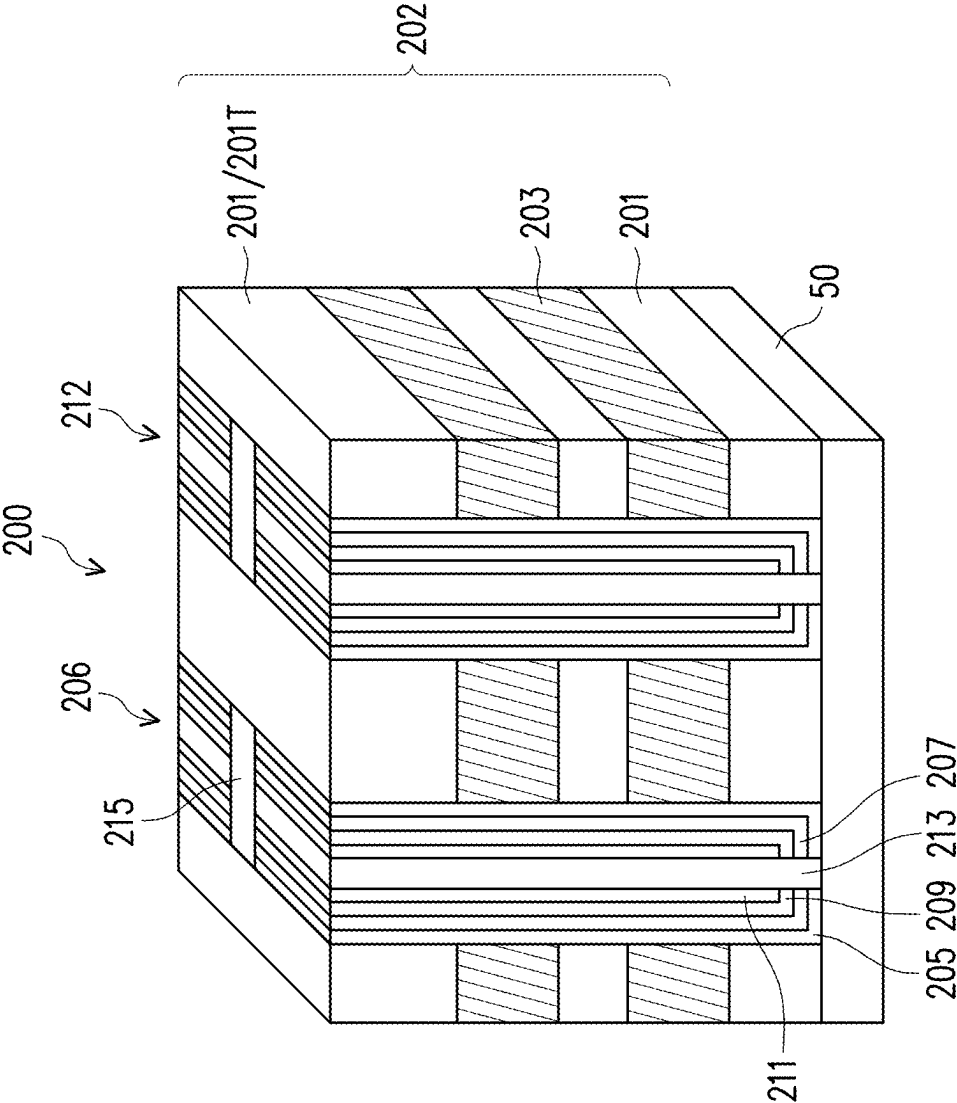


FIG. 10

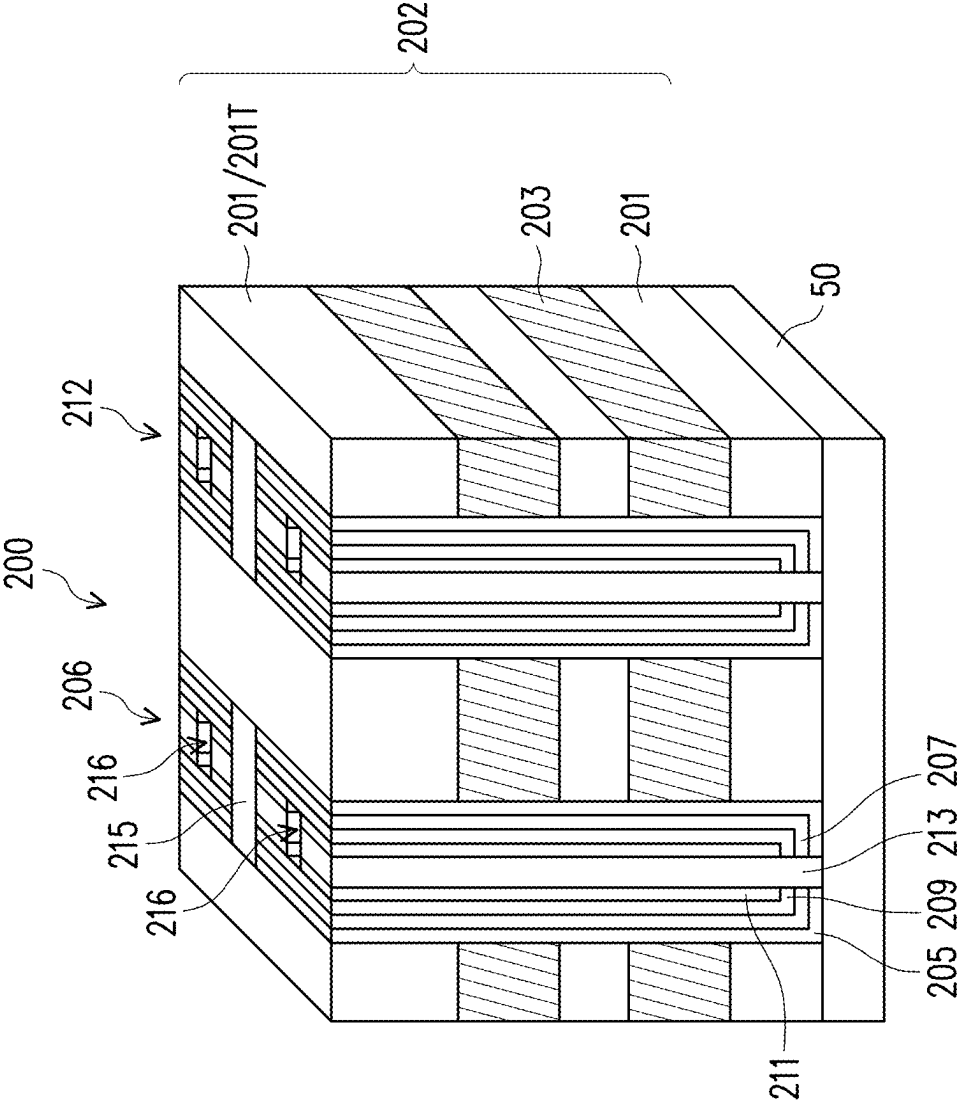


FIG. 11

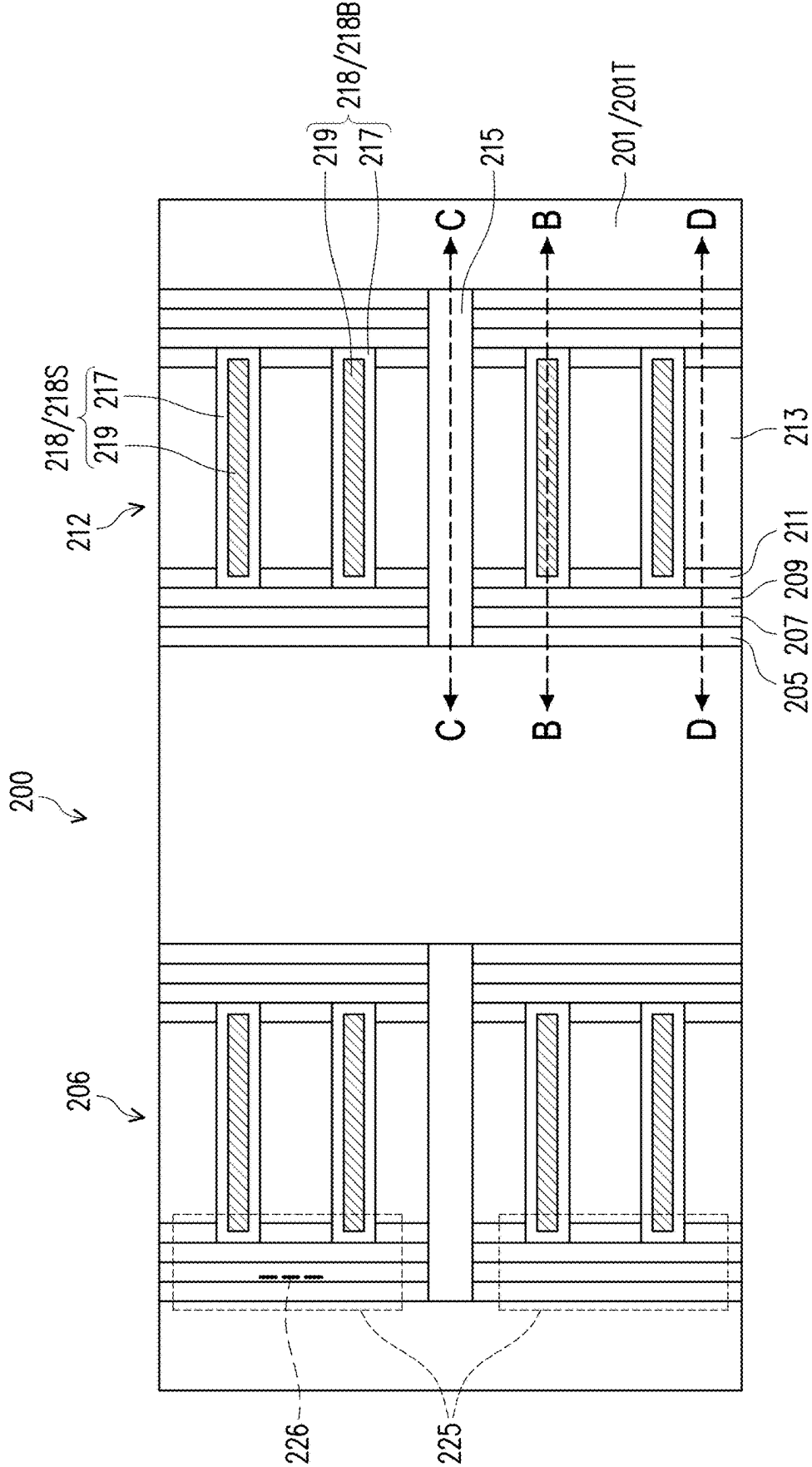


FIG. 12B

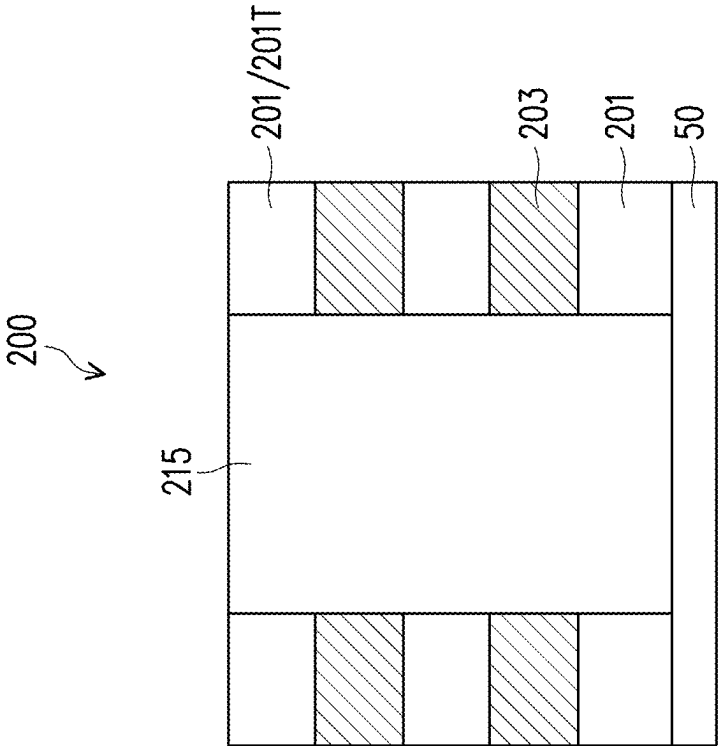


FIG. 12D

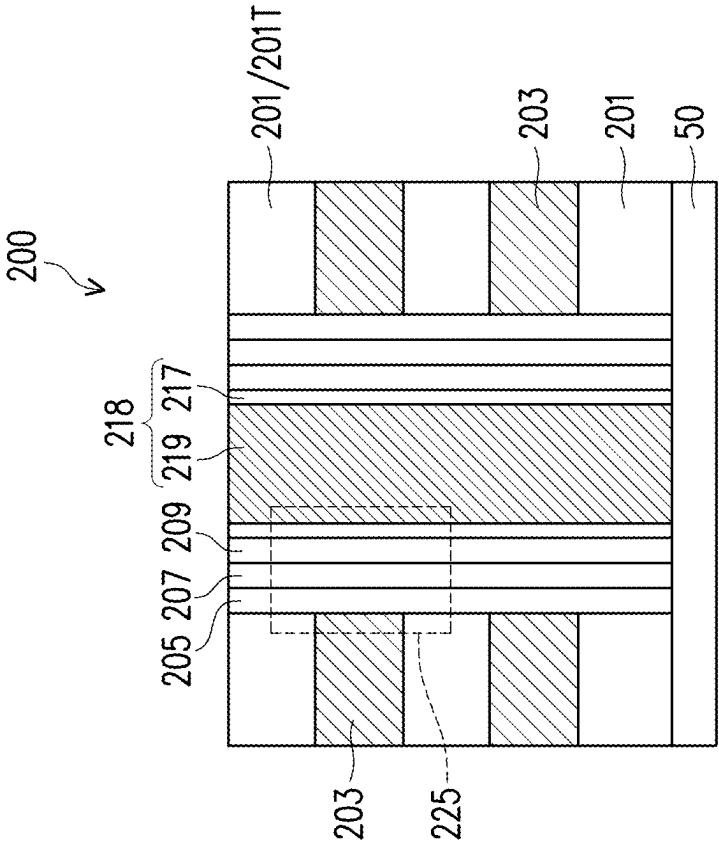


FIG. 12C

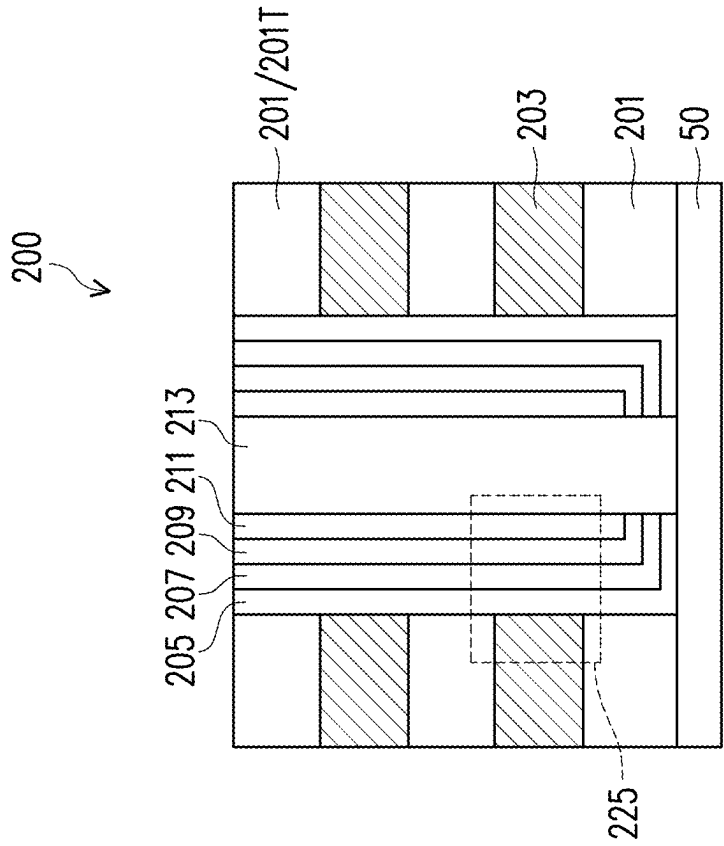


FIG. 12E

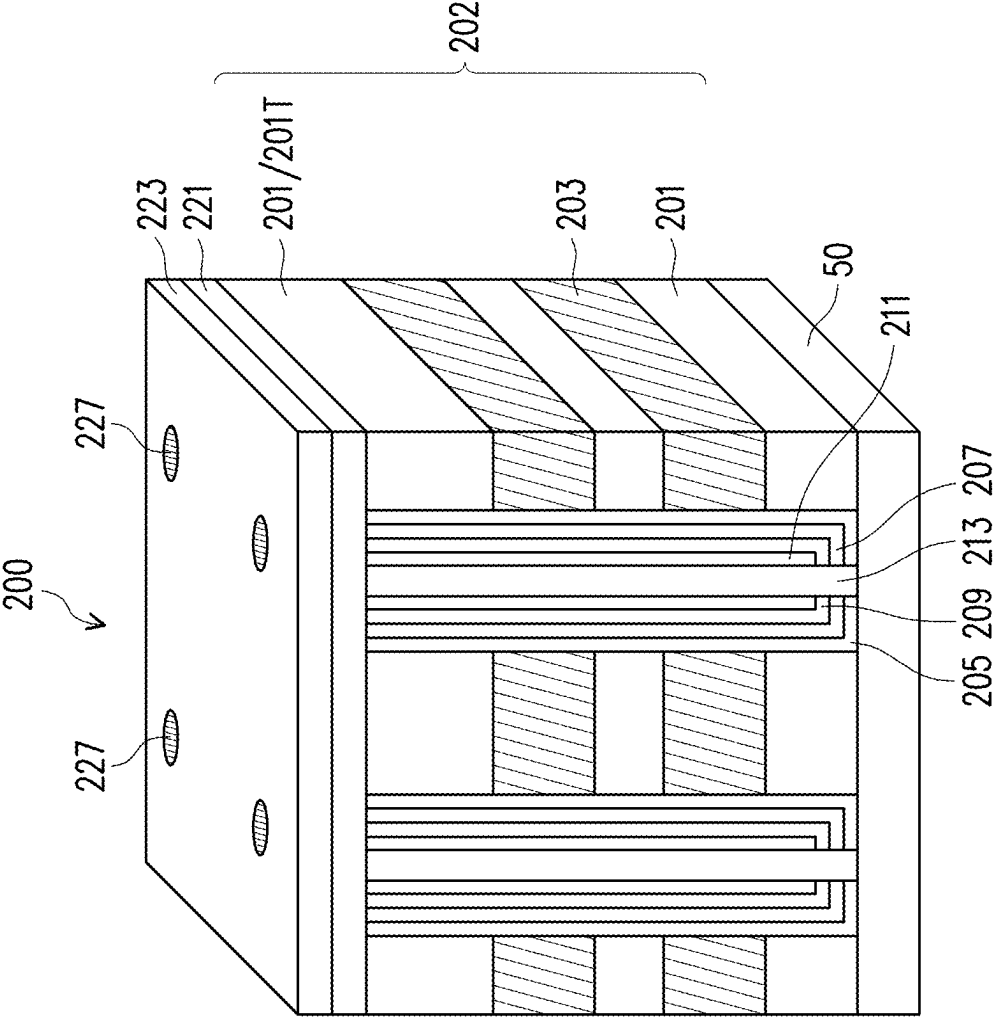


FIG. 13

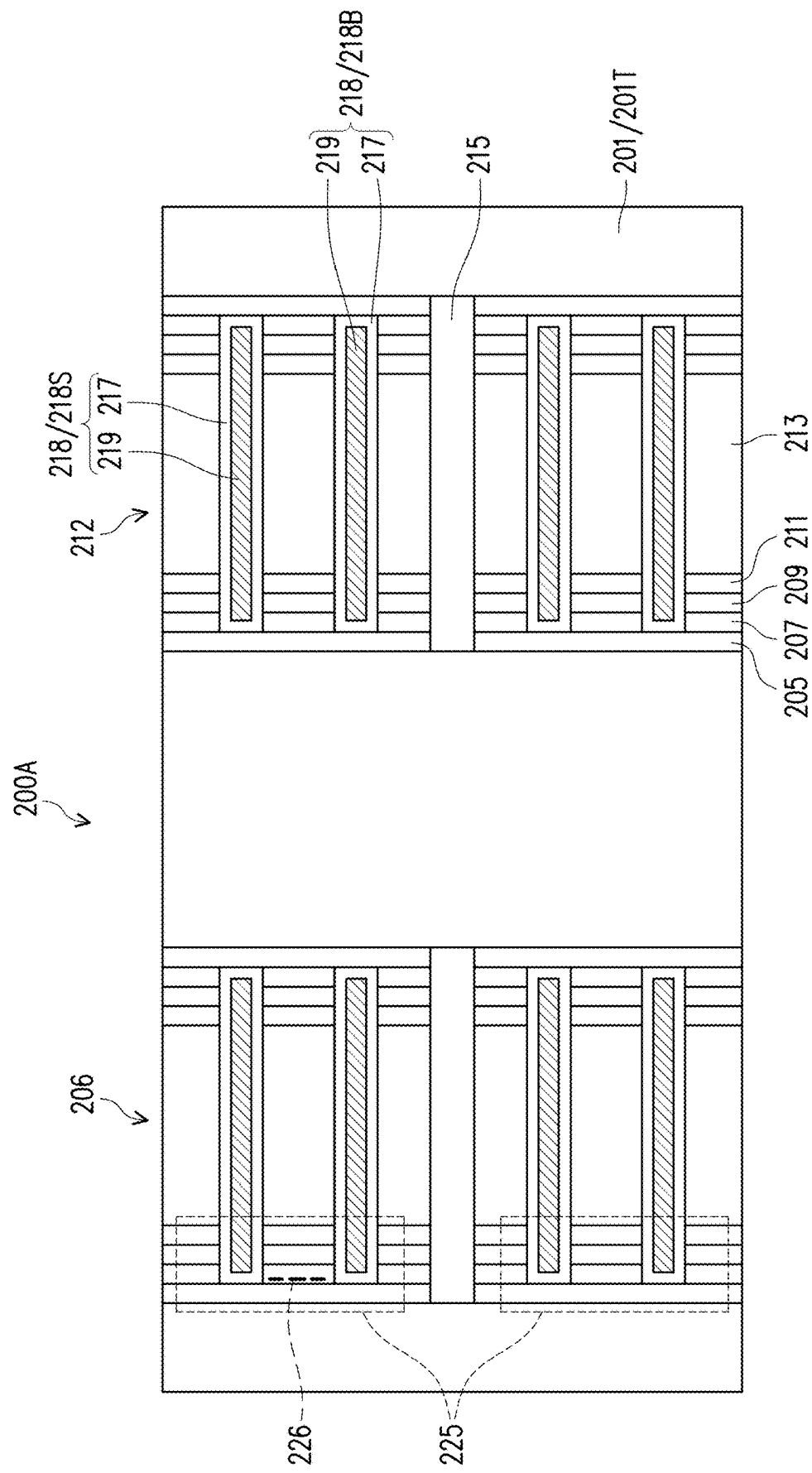


FIG. 14

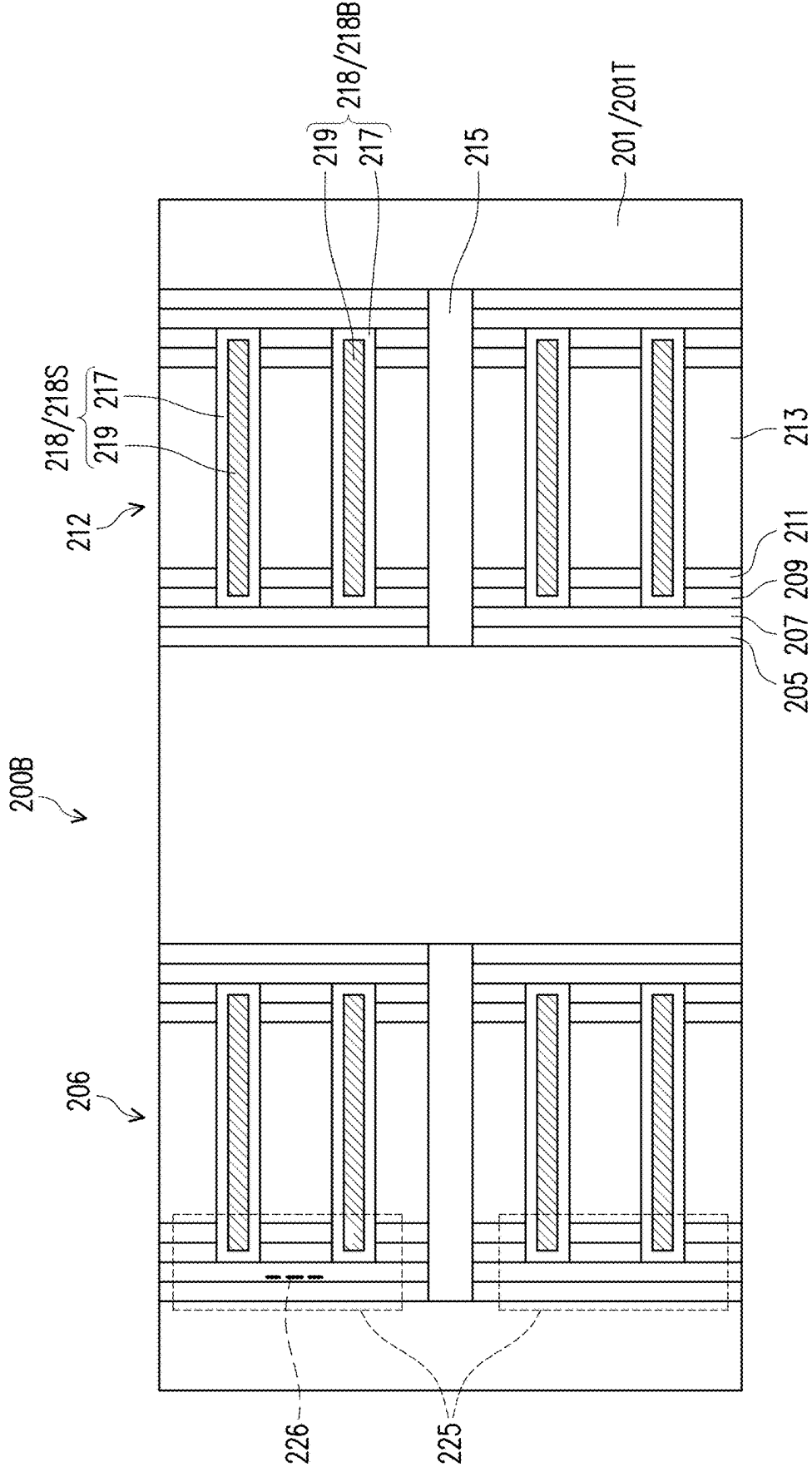


FIG. 15

1000

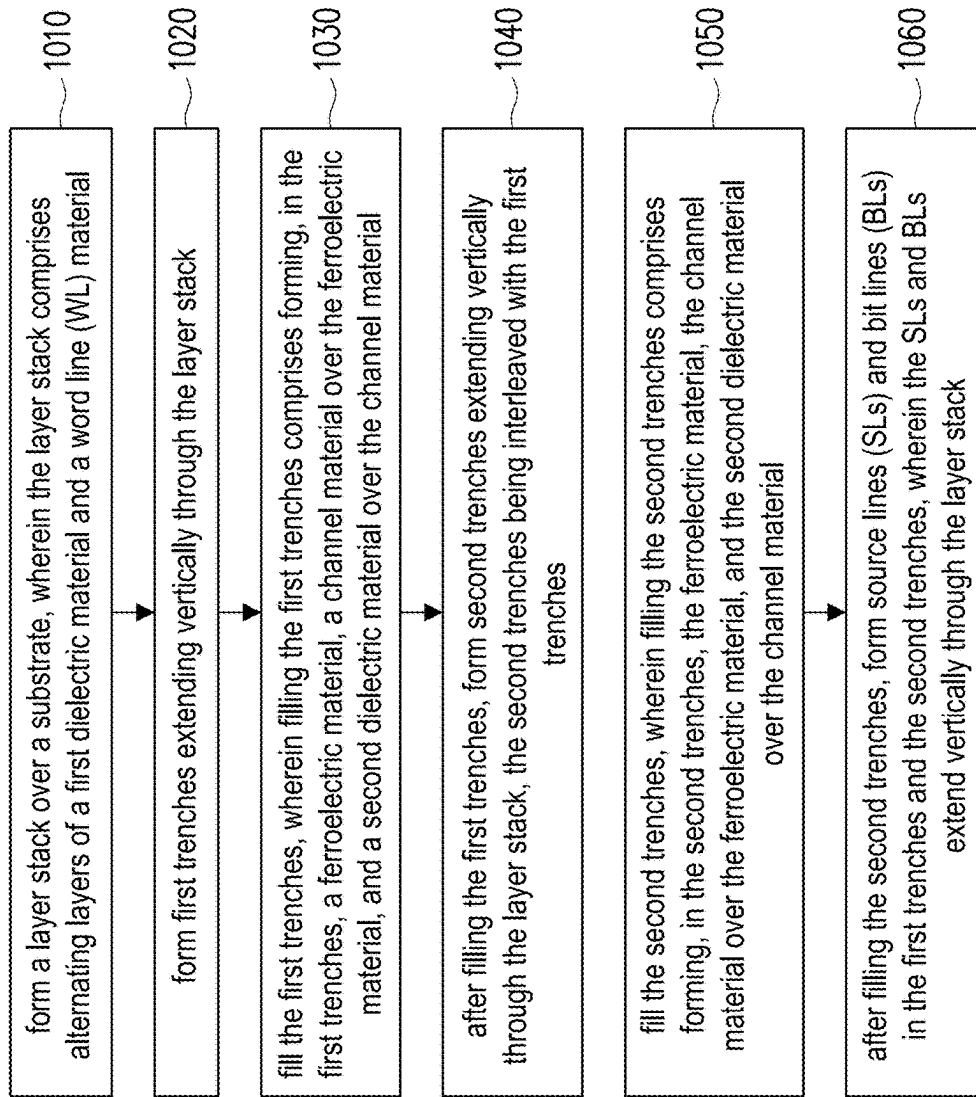


FIG. 16

THREE-DIMENSIONAL FERROELECTRIC RANDOM ACCESS MEMORY DEVICES AND METHODS OF FORMING

PRIORITY CLAIM AND CROSS-REFERENCE

[0001] This application is a continuation of U.S. patent application Ser. No. 18/413,668, filed Jan. 16, 2024 and entitled “Three-Dimensional Ferroelectric Random Access Memory Devices and Methods of Forming,” which is a continuation of U.S. patent application Ser. No. 17/316,167, filed May 10, 2021 and entitled “Three-Dimensional Ferroelectric Random Access Memory Devices and Methods of Forming” (now U.S. Pat. No. 11,903,214 issued Feb. 13, 2024), which claims the benefit of U.S. Provisional Application No. 63/052,499, filed on Jul. 16, 2020, which applications are hereby incorporated herein by reference.

BACKGROUND

[0002] Semiconductor memories are used in integrated circuits for electronic applications, including radios, televisions, cell phones, and personal computing devices, as examples. Semiconductor memories include two major categories. One is volatile memories; the other is non-volatile memories. Volatile memories include random access memory (RAM), which can be further divided into two sub-categories, static random access memory (SRAM) and dynamic random access memory (DRAM). Both SRAM and DRAM are volatile because they will lose the information stored when they are not powered.

[0003] On the other hand, non-volatile memories can keep data stored on them without power being supplied. One type of non-volatile semiconductor memory is ferroelectric random access memory (FeRAM, or FRAM). Advantages of FeRAM include its fast write/read speed and small size.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] For a more complete understanding of the present invention, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

[0005] FIG. 1 illustrates a cross-sectional view of a semiconductor device with integrated memory devices, in an embodiment;

[0006] FIGS. 2-8, 9A, 9B, 10, 11, 12A-12E, and 13 illustrate various views of a three-dimensional (3D) ferroelectric random access memory (FeRAM) device at various stages of manufacturing, in an embodiment;

[0007] FIG. 14 illustrates a top view of a three-dimensional (3D) ferroelectric random access memory (FeRAM) device, in another embodiment;

[0008] FIG. 15 illustrates a top view of a three-dimensional (3D) ferroelectric random access memory (FeRAM) device, in yet another embodiment; and

[0009] FIG. 16 illustrates a flow chart of a method of forming a three-dimensional (3D) ferroelectric random access memory (FeRAM) device, in some embodiments.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0010] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present

disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact.

[0011] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. Throughout the discussion herein, unless otherwise specified, the same or similar reference numeral in different figures refers to the same or similar element formed by a same or similar process using a same or similar material(s).

[0012] In some embodiments, a Litho-Etch-Dep-Litho-Etch-Dep (LEDLED) method is disclosed for forming high-density memory arrays in a ferroelectric random access memory (FeRAM) device. The LEDLED method includes forming a first plurality of trenches in a layer stack over a substrate, where the layer stack includes alternating layers of a first dielectric material and a word line (WL) material. The first plurality of trenches are then filled by forming a ferroelectric material, a channel material, and a second dielectric material successively in the first plurality of trenches. After filling the first plurality of trenches, a second plurality of trenches are formed in the layer stack, where the second plurality of trenches are interleaved with the first plurality of trenches. The second plurality of trenches are then filled with the ferroelectric material, the channel material, and the second dielectric material. Next, source lines (SLs) and bit lines (BLs) are formed in the first and second plurality of trenches. Compared with a reference method where the first and second plurality of trenches are formed simultaneously, the LEDLED method allows the first and second plurality of trenches to be formed closer to each other for higher integration density while avoiding the fin structure collapsing issue encountered by the reference method.

[0013] FIG. 1 illustrates a cross-sectional view of a semiconductor device 100 with integrated memory devices 123 (e.g., 123A and 123B), in an embodiment. The semiconductor device 100 is a fin-field effect transistor (FinFET) device with three-dimensional (3D) ferroelectric random access memory (FeRAM) devices 123 integrated in the back-end-of-line (BEOL) processing of semiconductor manufacturing, in the illustrated embodiment. To avoid clutter, details of the memory devices 123 are not shown in FIG. 1, but are illustrated in subsequent figures hereinafter. Note that FinFET is used as a non-limiting example here, the FeRAM devices 123 may be integrated in the BEOL processing with any suitable device, such as planar devices or gate-all-around (GAA) devices.

[0014] As illustrated in FIG. 1, the semiconductor device 100 includes different regions for forming different types of circuits. For example, the semiconductor device 100 may include a first region 110 for forming logic circuits, and may include a second region 120 for forming, e.g., peripheral

circuits, input/output (I/O) circuits, electrostatic discharge (ESD) circuits, and/or analog circuits. Other regions for forming other types of circuits are possible and are fully intended to be included within the scope of the present disclosure.

[0015] The semiconductor device **100** includes a substrate **101**. The substrate **101** may be a bulk substrate, such as a silicon substrate, doped or undoped, or an active layer of a semiconductor-on-insulator (SOI) substrate. The substrate **101** may include other semiconductor materials, such as germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, gallium nitride, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. Other substrates, such as multi-layered or gradient substrates, may also be used.

[0016] Electrical components, such as transistors, resistors, capacitors, inductors, diodes, or the like, are formed in or on the substrate **101** in the front-end-of-line (FEOL) processing of semiconductor manufacturing. In the example of FIG. **1**, semiconductor fins **103** (also referred to as fins) are formed protruding above the substrate **101**. Isolation regions **105**, such as shallow-trench isolation (STI) regions, are formed between or around the semiconductor fins **103**. Gate electrodes **109** are formed over the semiconductor fins **103**. Gate spacers **11** are formed along sidewalls of the gate electrodes **109**. Source/drain regions **107**, such as epitaxial source/drain regions, are formed on opposing sides of the gate electrodes **109**. Contacts **113**, such as gate contacts and source/drain contacts, are formed over and electrically coupled to respective underlying electrically conductive features (e.g., gate electrodes **109** or source/drain regions **107**). One or more dielectric layers **117**, such as an inter-layer dielectric (ILD) layer, is formed over the substrate **101** and around the semiconductor fins **103** and the gate electrodes **109**. Other electrically conductive features, such as interconnect structures comprising conductive lines **115** and vias **114**, may also be formed in the one or more dielectric layers **117**. The FinFETs in FIG. **1** may be formed by any suitable method known or used in the art, details are not repeated here. For ease of discussion herein, the substrate **101**, the electrical components (e.g., FinFETs) formed in or on the substrate **101**, the contacts **113**, conductive features **115/114**, and the one or more dielectric layers **117** are collectively referred to as substrate **50**.

[0017] Still referring to FIG. **1**, a dielectric layer **119**, which may be an etch stop layer (ESL), is formed over the one or more dielectric layers **117**. In an embodiment, the dielectric layer **119** is formed of silicon nitride using plasma-enhanced physical vapor deposition (PECVD), although other dielectric materials such as nitride, carbide, combinations thereof, or the like, and alternative techniques of forming the dielectric layer **119**, such as low-pressure chemical vapor deposition (LPCVD), PVD, or the like, could alternatively be used. In some embodiments, the dielectric layer **119** is omitted. Next, a dielectric layer **121** is formed over the dielectric layer **119**. The dielectric layer **121** may be any suitable dielectric material, such as silicon oxide, silicon nitride, or the like, formed by a suitable method, such as PVD, CVD, or the like. One or more memory device **123A**, each of which includes a plurality of memory cells, are formed in the dielectric layer **121** and coupled to electrically conductive features (e.g., vias **124**

and conductive lines **125**) in the dielectric layer **121**. Various embodiments of the memory devices **123A** or **123B** in FIG. **1** (e.g., 3D FeRAM devices **200**, **200A**, and **200B**) are discussed hereinafter in details.

[0018] FIG. **1** further illustrates a second layer of memory devices **123B** formed over the memory devices **123A**. The memory devices **123A** and **123B** may have a same or similar structure, and may be collectively referred to as memory devices **123**. The example of FIG. **1** illustrates two layers of memory devices **123** as a non-limiting example. Other numbers of layers of memory devices **123**, such as one layer, three layers, or more, are also possible and are fully intended to be included within the scope of the present disclosure. The one or more layers of memory device **123** are formed in a memory region **130** of the semiconductor device **100**, and may be formed in the back-end-of-line (BEOL) processing of semiconductor manufacturing. The memory devices **123** may be formed in the BEOL processing at any suitable locations within the semiconductor device **100**, such as over (e.g., directly over) the first region **110**, over the second region **120**, or over a plurality of regions.

[0019] In the example of FIG. **1**, the memory devices **123** occupy some, but not all, of the areas of the memory region **130** of the semiconductor device **100**, because other features, such as conductive lines **125** and vias **124**, may be formed in other areas of the memory region **130** for connection to conductive features over and below the memory region **130**. In some embodiments, to form the memory devices **123A** or **123B**, a mask layer, such as a patterned photoresist layer, is formed to cover some areas of the memory region **130**, while the memory devices **123A** or **123B** are formed in other areas of the memory region **130** exposed by the mask layer. After the memory devices **123** are formed, the mask layer is then removed.

[0020] Still referring to FIG. **1**, after the memory region **130** is formed, an interconnect structure **140**, which includes dielectric layer **121** and electrically conductive features (e.g., vias **124** and conductive lines **125**) in the dielectric layer **121**, is formed over the memory region **130**. The interconnect structure **140** may electrically connect the electrical components formed in/on the substrate **101** to form functional circuits. The interconnect structure **140** may also electrically couple the memory devices **123** to the components formed in/on the substrate **101**, and/or couple the memory devices **123** to conductive pads formed over the interconnect structure **140** for connection with an external circuit or an external device. Formation of interconnect structure is known in the art, thus details are not repeated here.

[0021] In some embodiments, the memory devices **123** are electrically coupled to the electrical components (e.g., transistors) formed on the substrate **50**, e.g., by the vias **124** and conductive lines **125**, and are controlled or accessed (e.g., written to or read from) by functional circuits of the semiconductor device **100**, in some embodiments. In addition, or alternatively, the memory devices **123** are electrically coupled to conductive pads formed over a top metal layer of the interconnect structure **140**, in which case the memory devices **123** may be controlled or accessed by an external circuit (e.g., another semiconductor device) directly without involvement of the functional circuits of the semiconductor device **100**, in some embodiments. Although additional metal layers (e.g., the interconnect structure **140**) are formed over the memory devices **123** in the example of FIG. **1**, the

memory devices **123** may be formed in a top (e.g., topmost) metal layer of the semiconductor device **100**, these and other variations are fully intended to be included within the scope of the present disclosure.

[0022] FIGS. 2-8, 9A, 9B, 10, 11, 12A-12E, and 13 illustrate various views (e.g., perspective view, cross-sectional view, top view) of a three-dimensional (3D) ferroelectric random access memory (FeRAM) device **200** at various stages of manufacturing, in an embodiment. For ease of discussion, a 3D FeRAM device may also be referred to as a 3D memory device, or simply a memory device in the discussion herein. The 3D memory device **200** is a three-dimensional memory device with a ferroelectric material. The 3D memory device **200** may be used as the memory device **123A** and/or **123B** in FIG. 1. Note that for simplicity, not all features of the 3D memory device **200** are illustrated in the figures, and the figures may only show a portion of the 3D memory device **200**.

[0023] Referring now to FIG. 2, which shows a perspective view of the memory device **200** at an early stage of fabrication. A layer stack **202** is formed over the substrate **50**. The layer stack **202** includes alternating layers of a dielectric material **201** and an electrically conductive material **203**. Each layer of the dielectric material **201** in the layer stack **202** may also be referred to as a dielectric layer **201**, and each layer of the electrically conductive material **203** in the layer stack **202** may also be referred to as an electrically conductive layer **203**. As will be discussed hereinafter, the electrically conductive material **203** is used to form word lines (WLs) of the 3D memory device **200**, and therefore, may also be referred to as word line material **203**.

[0024] In some embodiments, to form the layer stack **202**, the dielectric layer **201** is first formed by depositing a suitable dielectric material, such as silicon oxide, silicon nitride, or the like, on the substrate **50** using a suitable deposition method, such as PVD, CVD, atomic layer deposition (ALD), or the like. Next, the electrically conductive layer **203** is formed over the dielectric layer **201**. In some embodiments, the electrically conductive layer **203** is formed of an electrically conductive material, such as a metal or metal-containing material. Examples materials for the electrically conductive layer **203** include Al, Ti, TiN, TaN, Co, Ag, Au, Cu, Ni, Cr, Hf, Ru, W, Pt, or the like. The electrically conductive layer **203** may be formed by, e.g., PVD, CVD, ALD, combinations thereof, or the like. The above deposition processes repeat, until a target number of layers is formed in the layer stack **202**. In the illustrated embodiment, the topmost layer of the layer stack **202** is a dielectric layer **201**, which may be referred to as a topmost dielectric layer **201T** of the layer stack **202**. Note that the number of layers in the layer stack **202** can be any suitable number and is not limited to the example shown in FIG. 2.

[0025] Next, in FIG. 3, first trenches **206** (may also be referred to as openings, recesses, or slots) are formed in the layer stack **202** and extend through the layer stack **202** to expose the substrate **50**. The first trenches **206** may be formed using, e.g., photolithography and etching techniques. In the illustrated embodiment, the first trenches **206** extend from a lower surface of the layer stack **202** facing the substrate **50** to an upper surface of the layer stack **202** distal from the substrate **50**. In the example of FIG. 3, the first trenches **206** extends continuously between opposing sidewalls of the layer stack **202**, such that the first trenches **206** cut through the layer stack **202** and separate the layer stack

202 into a plurality of slices (e.g., fin shaped structures) that are separate (e.g., spaced apart) from each other. Note that for simplicity, FIG. 3 only shows one of the first trenches **206**. Referring temporarily to FIG. 9B, FIG. 9B shows a plurality of first trenches **206** interleaved with a plurality of second trenches **212** formed in the layer stack **202** in subsequent processing, where the first trenches **206** and second trenches **212** are filled with other materials (e.g., **205**, **207**, **209**, **211**, **213**) to form the 3D memory device **200**, details of which are discussed hereinafter.

[0026] Next, in FIG. 4, a ferroelectric material **205** is formed (e.g., conformally) in the first trenches **206** along sidewalls and bottoms of the first trenches **206**. The ferroelectric material **205** may also be formed over the upper surface of the layer stack **202**. Next, a channel material **207** is formed (e.g., conformally) over the ferroelectric material **205**. Next, a capping layer **209** is formed (e.g., conformally) over the channel material **207**, and an oxide liner **211** is formed (e.g., conformally) over the capping layer **209**.

[0027] In some embodiments, the ferroelectric material **205** comprises BaTiO₃, PbTiO₃, PbZrO₃, LiNbO₃, NaNbO₃, KNbO₃, KTaO₃, BiScO₃, BiFeO₃, Hf_{1-x}Er_xO, Hf_{1-x}La_xO, Hf_{1-x}Y_xO, Hf_{1-x}Gd_xO, Hf_{1-x}Al_xO, Hf_{1-x}Zr_xO, Hf_{1-x}Ti_xO, Hf_{1-x}Ta_xO, AlScN, the like, combinations thereof, or multi layers thereof, and may be formed by a suitable formation method such as PVD, CVD, ALD, or the like. The ferroelectric material **205** may also be referred to as ferroelectric film.

[0028] In some embodiments, the channel material **207** is a semiconductive material, such as amorphous-silicon (a-Si), polysilicon (poly-Si), a semiconductive oxide (e.g., indium gallium zinc oxide (IGZO), indium zinc oxide (IZO), zinc oxide (ZnO), indium tin oxide (ITO), or indium tungsten oxide (IWO)), or the like. The channel material **207** may be formed by, e.g., PVD, CVD, ALD, combinations thereof, or the like.

[0029] In some embodiments, the capping layer **209** is formed of a dielectric material. The capping layer **209** functions as a protection layer for the channel material **207** and prevents contaminant elements, such as H, Cl, or F in subsequent processing (e.g., a subsequent etching process), from diffusing into the channel material **207**. In an example embodiment, the capping layer **209** is formed of a high-k dielectric material with a dielectric constant (e.g., k-value) larger than, e.g., about 7.0 or even higher. Examples of high-k dielectric material include HfO₂, TiO₂, HfZrO, Ta₂O₃, HfSiO₄, ZrO₂, ZrSiO₂, and other suitable materials. The high-k dielectric material of the capping layer **209** may be formed by atomic layer deposition (ALD) and/or other suitable methods.

[0030] In some embodiments, the dielectric constant K_{cap} of the capping layer **209** (e.g., a dielectric material) is higher than the dielectric constant K_{fer} of the ferroelectric material **205** (e.g., $K_{cap} > K_{fer}$) to facilitate reliable switching of the electrical polarization direction of the ferroelectric material **205**. As will be discussed in more detail hereinafter, the electrical polarization direction of the ferroelectric material **205** is used to store the digital information (e.g., a bit of 0 or 1) of each memory cell of the 3D memory device **200**. The electrical polarization direction of the ferroelectric material **205** is switched by an electrical field applied to the ferroelectric material **205**, which electric field may be proportional to a voltage applied across the ferroelectric material **205**. Referring temporarily to FIG. 12C, during opera-

tion (e.g., a write operation) of a memory cell **225** of the 3D memory device **200**, a voltage V_{total} is applied between, e.g., a word line (WL) **203** and a respective source line (SL) **218**. The voltage V_{total} is shared by the different layers of materials between the WL **203** and the SL **218**, such as the ferroelectric material **205**, the channel material **207**, and the capping layer **209** in the example of FIG. 12C. In other words, a portion of the voltage V_{total} is applied to each layer (e.g., **205**, **207**, or **209**) of material between the WL **203** and the SL **218**. Since the voltage applied to each layer of material is inversely proportional to its dielectric constant, it may be advantageous to choose the dielectric constant K_{cap} of the capping layer **209** to be higher than the dielectric constant K_{fer} of the ferroelectric material **205**, such that a substantial percentage of the voltage V_{total} is applied across the ferroelectric material **205** to facilitate switching of the electrical polarization direction of the ferroelectric material **205**. For similar reasons, the dielectric constant of the channel material **207** is also chosen to be higher than the dielectric constant K_{fer} of the ferroelectric material **205**, in some embodiments. In some embodiments, the dielectric constant K_{cap} of the capping layer **209** is chosen to be twice, five times, ten times, or more than the dielectric constant K_{fer} of the ferroelectric material **205**. High dielectric constant materials, such as AlO_x (with k-value between about 9 and about 11), HfO_x (with k-value between about 25 and about 27), and TaO_2 (with k-value between about 40 and 80) may therefore be advantageously used as the material for the capping layer **209**, especially in applications where the voltage V_{total} is low. In other embodiments, the dielectric constant K_{cap} of the capping layer **209** may be substantially equal to, or smaller than, the dielectric constant K_{fer} of the ferroelectric material **205**, e.g., in applications where the voltage V_{total} is high enough such that the voltage applied across the ferroelectric material **205** is higher than the switching voltage for the ferroelectric material **205**.

[0031] Still referring to FIG. 4, the oxide liner **211** comprises an oxide, such as silicon oxide, and is formed by a suitable formation method, such as ALD, or the like, in some embodiments. The oxide liner **211** protects, e.g., sidewall portions of the ferroelectric material **205**, the channel material **207**, and the capping layer **209** in a subsequent etching process (see FIG. 5), in some embodiments.

[0032] Next, in FIG. 5, a suitable etching process, such as an anisotropic etching process, is performed to remove first portions of the oxide liner **211**, first portions of the capping layer **209**, first portions of the channel material **207**, and first portions of the ferroelectric material **205** from bottoms of the first trenches **206**, and as a result, the upper surface of the substrate **50** is exposed at the bottom of the first trenches **206**. In the illustrated embodiment, due to the anisotropy of the etching process, second portions of the oxide liner **211**, second portions of the capping layer **209**, second portions of channel material **207**, and second portions of the ferroelectric material **205** disposed between the layer stack **202** and sidewalls **211S** of the oxide liner **211** facing the first trench **206** remain after the anisotropic etching process. Note that the second portions of the capping layer **209**, the second portions of channel material **207**, and the second portions of the ferroelectric material **205** have L-shaped cross-sections, and have bottom portions that are disposed between the second portions of the oxide liner **211** and the substrate **50**. As illustrated in FIG. 5, each of the sidewalls **211S** of the oxide liner **211** is vertically aligned (e.g., in a same vertical

plane in FIG. 5) with respective sidewalls of the second portions of the capping layer **209**, the second portions of the channel material **207**, and the second portions of the ferroelectric material **205**.

[0033] Next, in FIG. 6, a dielectric material **213** is formed in the first trenches **206** to fill the remaining space in the first trenches **206**. In some embodiments, the dielectric material **213** is formed by depositing a suitable dielectric material, such as silicon oxide, silicon nitride, or the like, using a suitable deposition method, such as ALD, or the like. The dielectric material **213** may overfill the first trenches **206** and may be formed over the upper surface of the layer stack **202**. Next, a planarization process, such as chemical mechanical planarization (CMP), is performed to remove excess portions of the ferroelectric material **205**, the channel material **207**, the capping layer **209**, the oxide liner **211**, and the dielectric material **213** from the upper surface of the layer stack **202**. As illustrated in FIG. 6, after the planarization process, remaining portions of the ferroelectric material **205**, the channel material **207**, and the capping layer **209** have L-shaped cross-sections.

[0034] Next, in FIG. 7, second trenches **212** are formed in the layer stack **202**. The second trenches **212** are similar to the first trenches **206** and may be formed using a same or similar method. For simplicity, FIG. 7 shows only one second trench **212**. FIG. 9B illustrates a plurality of second trenches **212** in the layer stack **202** that are interleaved with the first trenches **206**. As illustrated in FIG. 9B, the first trenches **206** and the second trenches **212** are formed alternately in the layer stack **202**.

[0035] Next, in FIG. 8, the ferroelectric material **205**, the channel material **207**, the capping layer **209**, and the oxide liner **211** are formed (e.g., conformally) successively in the second trenches **212**. The materials and the formation methods of the ferroelectric material **205**, the channel material **207**, the capping layer **209**, and the oxide liner **211** may be the same as or similar to those discussed above with reference to FIG. 4, thus details are not repeated.

[0036] Next, in FIG. 9A, a suitable etching process, such as an anisotropic etching process, is performed to remove portions of the dielectric material **213**, portions of the oxide liner **211**, portions of the capping layer **209**, portions of the channel material **207**, and portions of the ferroelectric material **205** from bottoms of the second trenches **212**. Details of the etching process may be the same as or similar to those discussed above with reference to FIGS. 5, thus details are not repeated.

[0037] Next, the dielectric material **213** is formed in the second trenches **212** and completely fills the second trenches **212**. A planarization process, such as CMP, is performed next to remove excess portions of the oxide liner **211**, the capping layer **209**, the channel material **207**, and the ferroelectric material **205** from the upper surface of the layer stack **202**.

[0038] FIG. 9B illustrates a cross-sectional view of the 3D memory device **200** of FIG. 9A along cross-section A-A. Note that FIG. 9B illustrates a larger portion of the 3D memory device **200** than FIG. 9A, and therefore, shows a plurality of first trenches **206** interleaved with a plurality of second trenches **212**. The first trenches **206** and the second trenches **212** in FIG. 9B are filled with various layers of materials (e.g., **205**, **207**, **209**, **211**, and **213**) for forming memory cells in subsequent processing. The first trenches

206 and the second trenches 212 may be collectively referred to as trenches 206/212.

[0039] The disclosed method forms the structure of FIGS. 9A and 9B by forming first trenches 206 (e.g., using lithography and etching techniques), filling the trenches 206 (e.g., by deposition of layers of materials 205/207/209/211/213), forming the second trenches 212 (e.g., using lithography and etching techniques), and filling the second trenches 212 (e.g., by deposition of layers of materials 205/207/209/211/213). Therefore, the disclosed method may also be referred to as a Litho-Etch-Dep-Litho-Etch-Dep (LEDLED) method. Compared with a reference method where the first trenches 206 and the second trenches 212 are formed at the same time (e.g., by a same etching process), the disclosed LEDLED method allows trenches 206/212 to be formed closer to each other without suffering from the “fin collapsing” issue discussed below, thereby enabling memory cells to be formed in higher density than otherwise achievable.

[0040] As semiconductor manufacturing process continues to advance, feature size continues to shrink in order to achieve higher integration density. To increase the memory cell density of the 3D memory device 200, it may be advantageous to form the layer stack 202 with a large number of layers (e.g., 201, 203), and to form the trenches 206/212 with smaller distance W1 between adjacent ones of the trenches 206/212. If the trenches 206/212 are formed at the same time, then the portions of layer stack 202 disposed between adjacent ones of the trenches 206/212 (which may be referred to as fin shaped structures of the layer stack 202) have a high aspect ratio of $H/W1$, where H is the height of the layer stack 202. Fin shaped structures of the layer stack 202 with high aspect ratio may collapse during subsequent processing, causing defects in the 3D memory device 200. In contrast, by using the disclosed LEDLED method, the aspect ratio of the fin shaped structures when forming the first trenches 206 is $H/W2$, which is much lower than $H/W1$, because the distance W2 between adjacent ones of the first trenches 206 is larger than W1. Similarly, when forming the second trenches 212, the first trenches 206 are already filled, the aspect ratio of the fin shaped structure (which includes the filled trenches 206) is $H/W3$ and is much lower than $H/W1$. As a result, the fin collapsing issue related with high aspect ratio is reduced or avoided by the disclosed LEDLED method, and device reliability and production yield are improved.

[0041] Next, in FIG. 10, isolation regions 215 are formed in the trenches 206/212 and extend vertically through the layer stack 202. The isolation regions 215 may also be referred to as memory cell isolation regions 215, or dielectric plugs 215. In some embodiments, to form the isolation regions 215, a patterned mask is formed over the upper surface of the layer stack 202, where patterns (e.g., openings) of the patterned mask layer correspond to locations of the isolation regions 215. Next, an anisotropic etching process is performed using the patterned mask layer as an etching mask to form openings in the layer stack 202 that extend vertically through the layer stack 202. Next, the openings in the layer stack 202 are filled with a dielectric material, such as silicon oxide, silicon nitride, or the like, using a suitable formation method such as CVD, PVD, ALD, or the like. A planarization process, such as CMP, may be performed next to remove excessive portions of the dielectric material from the upper surface of the layer stack 202,

and remaining portions of the dielectric material in the openings form the isolation regions 215.

[0042] In the example of FIG. 10, a width of the isolation regions 215 is substantially the same as a width of a respective trench 206/212 that the isolation region 215 is in, such that each isolation region 215 in the respective trench 206/212 physically contacts sidewalls of the layer stack 202 facing the respective trench. In other embodiments, the width of the isolation region 215 may be smaller than the width of the trench 206/212, and each isolation region 215 in a trench 206/212 extends continuously between sidewalls of the ferroelectric material 205 in the trench. In other words, each isolation region 215 physically contacts opposing sidewalls of the ferroelectric material 205 in the trench, but does not extend through the ferroelectric material 205.

[0043] Next, in FIG. 11, openings 216 are formed (e.g., by photolithography and etching techniques) in the oxide liner 211 and the dielectric material 213, which openings 216 extend from the upper surface of the layer stack 202 facing away from the substrate 50 to the lower surface of the layer stack 202 facing the substrate 50. The openings 216 expose sidewalls of the capping layer 209, in the example of FIG. 11. The openings 216 are filled with electrically conductive material(s) (e.g., 217, 219) to form source lines (SLs) 218S and bit lines (BLs) 218B in subsequent processing. In other embodiments, the openings 216 are formed to be wider than the opening 216 of FIG. 11, and exposes the sidewalls of the channel material 207 (see FIG. 15), or exposes the sidewalls of the ferroelectric material 205 (see FIG. 14).

[0044] Next, in FIG. 12A, a barrier layer 217 is formed (e.g., conformally) in the openings 216 to line sidewalls and bottoms of the openings 216. The barrier layer 217 may comprise titanium nitride, although other suitable material, such as tantalum nitride, titanium, tantalum, or the like, may also be used. A suitable formation method, such as CVD, ALD, or the like, may be performed to form the barrier layer 217. Next, an anisotropic etching process, which is optional, is performed to remove portions of the barrier layer 217 from the bottoms of the openings 216, such that the substrate 50 is exposed at the bottom of the openings 216. Next, an electrically conductive material 219, such as Al, Ti, TiN, TaN, Co, Ag, Au, Cu, Ni, Cr, Hf, Ru, W, Pt, or the like, is formed to fill the openings 216. A planarization process, such as CMP, may be performed next to remove excess portions of the barrier layer 217 and excess portions of the electrically conductive material 219 from the upper surface of the layer stack 202. Remaining portions of the barrier layer 217 and remaining portions of the electrically conductive material 219 in the openings 216 form conductive lines 218. The conductive lines 218 are metal columns or metal pillar that extends vertically through the layer stack 202, in the illustrated embodiments. The conductive lines 218 may also be referred to as the source lines (SLs) 218S or bit lines (BLs) 218B of the 3D memory device 200.

[0045] FIG. 12B illustrates a top view of the memory device 200 of FIG. 12A. FIGS. 12C, 12D, and 12E illustrate cross-sectional views of the memory device 200 of FIG. 12B along cross-sections B-B, C-C, and D-D, respectively. Note that for clarity, FIGS. 12B, 12C, 12D, and 12E may illustrate a portion of the 3D memory device 200 that is larger or smaller than the portion of 3D memory device 200 illustrated in FIG. 12A.

[0046] As illustrated in the top view of FIG. 12B, each isolation region 215 extends continuously from a first side-

wall of the layer stack **202** (e.g., a first sidewall of the topmost dielectric layer **201T** of the layer stack **202**) to a second sidewall of the layer stack **202** (e.g., a second sidewall of the topmost dielectric layer **201T** of the layer stack **202**) facing the first sidewall of the layer stack **202**, where the first sidewall and the second sidewall of the layer stack **202** are sidewalls of the layer stack **202** exposed by a same trench **206/212**. In other words, a width of the isolation region **215**, measured along the horizontal direction of FIG. **12B**, is the same as a distance between inner sidewalls of the layer stack **202** exposed by a same trench and facing each other. In addition, each of the conductive lines **218** extends continuously from a first sidewall of the capping layer **209** to a second sidewall of the capping layer **209** facing the first sidewall of the capping layer **209**. In other words, a width of the conductive line **218**, measured along the horizontal direction of FIG. **12B**, is the same as a distance between inner sidewalls of the capping layer **209** in a trench and facing each other.

[0047] In FIG. **12B**, a few, but not all, of the memory cells **225** of the 3D memory device **200** are highlighted by dashed boxes. Memory cells **225** are also highlighted by dashed boxes in FIGS. **12C** and **12E**. As illustrated in FIGS. **12A-12E**, each memory cell **225** is a transistor with an embedded ferroelectric film **205**. Within each memory cell **225**, the electrically conductive layer **203** (see, e.g., FIG. **12C**) functions as the gate electrode of the transistor, the conductive lines **218S** and **218B** (see, e.g., FIG. **12B**) function as the source/drain regions of the transistor, and the channel material **207** functions as the channel layer between the source/drain regions. The dashed line **226** in FIG. **12B** illustrates the channel region formed in the channel material **207** during operation of the 3D memory device **200**, e.g., when a voltage is applied at the gate of the transistor and causes the transistor to be turned on. The electrical polarization direction of the ferroelectric film **205** in each memory cell **225** indicates the digital information (e.g., a “0” or “1”) stored in the memory cell **225**, and determines the threshold voltage of the transistor of the memory cell **225**, more details are discussed hereinafter.

[0048] In the context of memory devices, the electrically conductive layer **203** (e.g., the gate electrode) in each memory cell **225** is referred to as the word line (WL) of the memory cell, the conductive lines **218S** and **218B** (e.g., the source/drain regions) may be referred to as the source line (SL) and the bit line (BL) of the memory cell.

[0049] As illustrated in FIG. **12A**, each of the electrically conductive layers **203** (e.g., WL) of the memory device **200** electrically connects multiple memory cells formed along a same horizontal plane (e.g., at a same vertical distance from the substrate **50**). In addition, as illustrated in FIG. **12C**, each SL or BL **218** electrically connects multiple vertically stacked memory cells **225**. Therefore, the disclosed 3D memory device **200** achieves efficient sharing of the WLs, BLs, and SLs among multiple memory cells **225**, and the 3D structure of the memory cells **225** allows for multiple layers of the memory cells **225** to be stacked easily together to form high density memory arrays.

[0050] FIGS. **12C** and **12E** illustrate different shapes for the cross-sections of the ferroelectric material **205**, the channel material **207**, and the capping layer **209** at different locations of the 3D memory device **200**. For example, in FIG. **12E**, which illustrates the cross-section along a vertical plane (vertical to the upper surface of the substrate **50**)

cutting across the dielectric material **213** (e.g., along cross-section D-D in FIG. **12B**), the ferroelectric material **205**, the channel material **207**, and the capping layer **209** have L-shaped cross-sections. In FIG. **12C**, which illustrates the cross-section along a vertical plane (vertical to the upper surface of the substrate **50**) cutting across the conductive lines **218** (e.g., along cross-section B-B in FIG. **12B**), the ferroelectric material **205**, the channel material **207**, and the capping layer **209** have rectangular shaped cross-sections.

[0051] Referring to FIGS. **12A-12E**, to perform a write operation on a particular memory cell **225**, a write voltage is applied across a portion of the ferroelectric material **205** within the memory cell **225**. The write voltage may be applied, for example, by applying a first voltage to the gate electrode **203** of the memory cell **225**, and applying a second voltage to the source/drain regions **218S/218B**. The voltage difference between the first voltage and the second voltage sets the polarization direction of the ferroelectric material **205**. Depending on the polarization direction of the ferroelectric material **205**, the threshold voltage V_T of the corresponding transistor of the memory cell **225** can be switched from a low threshold voltage V_L to a high threshold voltage V_H , or vice versa. The threshold voltage value (V_L or V_H) of the transistor can be used to indicate a bit of “0” or a “1” stored in the memory cell.

[0052] To perform a read operation on the memory cell **225**, a read voltage, which is a voltage between the low threshold voltage V_L and the high threshold voltage V_H , is applied to the gate electrode **203**. Depending on the polarization direction of the ferroelectric material **205** (or the threshold voltage V_T of the transistor), the transistor of the memory cells **225** may or may not be turned on. As a result, when a voltage is applied, e.g., between the source/drain regions **218S** and **218B**, an electrical current may or may not flow between the source/drain regions **218S** and **218B**. The electrical current may thus be detected to determine the digital bit stored in the memory cell.

[0053] Next, in FIG. **13**, contacts **227** are formed over the layer stack **202** to electrically couple to the conductive lines **218**. To form the contacts **227**, dielectric layers **221** and **223** are formed over the layer stack **202**. Openings are formed in the dielectric layers **221** and **223** to expose underlying conductive lines **218**. An electrically conductive material(s) is formed in the openings to form the contacts **227**.

[0054] In some embodiments, the dielectric layer **221** is formed of silicon nitride, although other suitable dielectric layer, such as silicon oxynitride, silicon carbide, may also be used. The dielectric layer **221** may be formed by, e.g., CVD, ALD, or the like. The dielectric layer **223** may be formed of a dielectric material such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate Glass (BPSG), undoped silicate glass (USG), or the like, and may be deposited by any suitable method, such as CVD, PECVD, or FCVD. The openings in the dielectric layers **221** and **223** may be formed by photolithography and etching techniques. The electrically conductive material(s) filling the opening may be any suitable conductive material, such as Al, Ti, TiN, TaN, Co, Ag, Au, Cu, Ni, Cr, Hf, Ru, W, Pt, or the like.

[0055] Additional processing may be performed to finish the 3D memory device **200**, as skilled artisans readily appreciate. For example, contacts may be formed to electrically couple to the WLs **203**, and portions of the layer stack **202** may be removed to form a stair-case shaped region

to facilitate formation of the contacts coupled to the WLs. For simplicity, details are not discussed here.

[0056] FIG. 14 illustrates a top view of a three-dimensional (3D) ferroelectric random access memory (FeRAM) device 200A, in another embodiment. The 3D memory device 200A is similar to the 3D memory device 200 of FIG. 12B and may be formed by similar processing steps, but the SL 218S and BLs 218B are wider than those in FIG. 12B and extend to (e.g., physically contact) sidewalls of the ferroelectric material 205 in a respective trench 206/212.

[0057] FIG. 15 illustrates a top view of a three-dimensional (3D) ferroelectric random access memory (FeRAM) device 200B, in yet another embodiment. The 3D memory device 200B is similar to the 3D memory device 200 of FIG. 12B and may be formed by similar processing steps, but the SL 218S and BLs 218B in FIG. 15 are wider than those in FIG. 12B and narrower than those in FIG. 14, and extend to (e.g., physically contact) sidewalls of the channel material 207 in a respective trench 206/212.

[0058] Embodiments may achieve advantages. For example, the disclosed LEDLED method, by forming first trenches 206 and the second trenches 212 in different processing steps, avoids or reduces the fin structure collapsing problem. As a result, device reliability and production yield are improved.

[0059] FIG. 16 illustrates a flow chart of a method 1000 of forming a three-dimensional (3D) ferroelectric random access memory (FeRAM) device, in some embodiments. It should be understood that the embodiment method shown in FIG. 16 is merely an example of many possible embodiment methods. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. For example, various steps as illustrated in FIG. 16 may be added, removed, replaced, rearranged, or repeated.

[0060] Referring to FIG. 16, at block 1010, a layer stack is formed over a substrate, wherein the layer stack comprises alternating layers of a first dielectric material and a word line (WL) material. At block 1020, first trenches are formed to extend vertically through the layer stack. At block 1030, the first trenches are filled, wherein filling the first trenches comprises forming, in the first trenches, a ferroelectric material, a channel material over the ferroelectric material, and a second dielectric material over the channel material. At block 1040, after filling the first trenches, second trenches are formed to extend vertically through the layer stack, the second trenches being interleaved with the first trenches. At block 1050, the second trenches are filled, wherein filling the second trenches comprises forming, in the second trenches, the ferroelectric material, the channel material over the ferroelectric material, and the second dielectric material over the channel material. At block 1060, after filling the second trenches, source lines (SLs) and bit lines (BLs) are formed in the first trenches and the second trenches that extend vertically through the layer stack.

[0061] In accordance with an embodiment, a method of forming a ferroelectric random access memory (FeRAM) device includes: forming a layer stack over a substrate, wherein the layer stack comprises alternating layers of a first dielectric material and a word line (WL) material; forming first trenches extending vertically through the layer stack from an upper surface of the layer stack distal from the substrate to a lower surface of the layer stack facing the substrate; lining bottoms and sidewalls of the first trenches with a ferroelectric material; forming a channel material in

the first trenches over the ferroelectric material; filling the first trenches with a second dielectric material; after filling the first trenches, forming second trenches extending vertically through the layer stack, wherein the second trenches are interleaved with the first trenches; lining bottoms and sidewalls of the second trenches with the ferroelectric material; forming the channel material in the second trenches over the ferroelectric material; filling the second trenches with the second dielectric material; and after filling the second trenches, forming source lines (SLs) and bit lines (BLs) in the first trenches and the second trenches, wherein the SLs and BLs extend vertically through the layer stack. In an embodiment, the WL material is an electrically conductive material, and the channel material is a semiconductive oxide. In an embodiment, the channel material comprises indium gallium zinc oxide, indium zinc oxide, zinc oxide, indium tin oxide, or indium tungsten oxide. In an embodiment, the method further includes: after forming the channel material in the first trenches and before filling the first trenches, forming a capping layer in the first trenches over the channel material, wherein the capping layer is a dielectric material, wherein a dielectric constant of the capping layer is higher than a dielectric constant of the ferroelectric material; and after forming the channel material in the second trenches and before filling the second trenches, forming the capping layer in the second trenches over the channel material. In an embodiment, the capping layer is formed of a high-k dielectric material. In an embodiment, forming the SLs and BLs comprises: forming openings in the first trenches and the second trenches, wherein the openings extend vertically through the layer stack; forming a barrier layer along sidewalls of the openings; and filling the openings with an electrically conductive material. In an embodiment, the SLs and the BLs extend continuously between, and contact, opposing sidewalls of the capping layer facing each other. In an embodiment, the SLs and the BLs extend continuously between, and contact, opposing sidewalls of the channel material facing each other. In an embodiment, the SLs and the BLs extend continuously between, and contact, opposing sidewalls of the ferroelectric material facing each other. In an embodiment, the method further includes: after forming the capping layer in the first trenches and before filling the first trenches, forming an oxide liner in the first trenches over the capping layer. In an embodiment, the method further includes: after forming the oxide liner in the first trenches and before filling the first trenches, performing an anisotropic etching process to remove portions of the oxide liner, portions of the capping layer, portions of the channel material, and portions of the ferroelectric material from the bottoms of the first trenches. In an embodiment, the method further includes forming isolation regions in the first trenches and in the second trenches between the BLs and the SLs, wherein the isolation regions extend vertically through the layer stack, and wherein in a top view, each of the isolation regions extends through the ferroelectric material in a respective trench of the first trenches and the second trenches.

[0062] In accordance with an embodiment, a method of forming a ferroelectric random access memory (FeRAM) device includes: forming a layer stack over a substrate, wherein the layer stack comprises alternating layers of a first dielectric material and a word line (WL) material; forming first trenches extending vertically through the layer stack; filling the first trenches, wherein filling the first trenches

comprises forming, in the first trenches, a ferroelectric material, a channel material over the ferroelectric material, and a second dielectric material over the channel material; after filling the first trenches, forming second trenches extending vertically through the layer stack, the second trenches being interleaved with the first trenches; and filling the second trenches, wherein filling the second trenches comprises forming, in the second trenches, the ferroelectric material, the channel material over the ferroelectric material, and the second dielectric material over the channel material. In an embodiment, filling the first trenches further comprises forming a capping layer in the first trenches between the channel material and the second dielectric material, wherein the capping layer is formed of a third dielectric material having a dielectric constant higher than that of the ferroelectric material. In an embodiment, filling the first trenches further comprises: forming an oxide liner in the first trenches between the capping layer and the second dielectric material; and before forming the second dielectric material in the first trenches, removing portions of the oxide liner, portions of the capping layer, portions of the channel material, and portions of the ferroelectric material from the bottoms of the first trenches by performing an anisotropic etching process. In an embodiment, the method further includes, after filling the second trenches, forming source lines (SLs) and bit lines (BLs) in the first trenches and the second trenches, wherein the SLs and BLs extend vertically through the layer stack, wherein forming the SLs and BLs comprises: forming openings in the first trenches and the second trenches, the openings extending vertically through the layer stack; lining sidewalls and bottoms of the openings with a barrier layer; and filling the openings with an electrically conductive material. In an embodiment, the method further includes forming isolation regions between the SLs and BLs, wherein forming the isolation regions comprises: forming openings in the second dielectric material; and filling the openings with a third dielectric material, the third dielectric material in the openings forming the isolation regions, wherein each of the isolation regions extends continuously from a first sidewall of a respective trench to a second opposing sidewall of the respective trench.

[0063] In accordance with an embodiment, a ferroelectric random access memory device includes: a layer stack over a substrate, the layer stack comprising alternating layers of a first dielectric material and a word line (WL) material; a dielectric layer embedded in the layer stack and extending from an upper surface of the layer stack distal to the substrate to a lower surface of the layer stack facing the substrate; a ferroelectric film between the dielectric layer and the layer stack, the ferroelectric film disposed on opposing sides of the dielectric layer; a channel layer between the ferroelectric film and the dielectric layer; a capping layer between the channel layer and the dielectric layer, wherein the capping layer is a dielectric material having a first dielectric constant higher than a second dielectric constant of the ferroelectric film; and source lines (SLs) and bit lines (BLs) embedded at least partially in the dielectric layer and extending through the layer stack. In an embodiment, the WL material is an electrically conductive material, and the channel layer is a semiconductive oxide. In an embodiment, the ferroelectric random access memory device further includes an oxide liner between the capping layer and the dielectric layer, wherein a first portion of the ferroelectric film, a second portion of the channel layer, and a third

portion of the capping layer are disposed between the oxide liner and the substrate, wherein a sidewall of the oxide liner facing the dielectric layer is in a same plane with a first sidewall of the first portion of the ferroelectric film, a second sidewall of the second portion of the channel layer, and a third sidewall of the third portion of the capping layer.

[0064] While this invention has been described with reference to illustrative embodiments, this description is not intended to be construed in a limiting sense. Various modifications and combinations of the illustrative embodiments, as well as other embodiments of the invention, will be apparent to persons skilled in the art upon reference to the description. It is therefore intended that the appended claims encompass any such modifications or embodiments.

What is claimed is:

1. A method of forming a ferroelectric random access memory (FeRAM) device, the method comprising:

forming a layer stack over a substrate by forming alternating layers of a first dielectric material and a first electrically conductive material over the substrate; forming first trenches extending vertically through the layer stack;

filling the first trenches, wherein filling the first trenches comprises forming a ferroelectric material, a channel material, and a second dielectric material in the first trenches;

after filling the first trenches, forming second trenches extending vertically through the layer stack, wherein locations of the second trenches are interleaved with locations of the first trenches;

filling the second trenches, wherein filling the second trenches comprises forming the ferroelectric material, the channel material, and the second dielectric material in the second trenches; and

after filling the second trenches, forming a source line (SL) and a bit line (BL) that are embedded in the second dielectric material, wherein the SL and the BL extend vertically through the layer stack.

2. The method of claim 1, wherein filling the first trenches comprises:

lining sidewalls and bottoms of the first trenches with the ferroelectric material;

forming the channel material over the ferroelectric material; and

forming the second dielectric material over the channel material.

3. The method of claim 2, further comprising, after forming the channel material and before forming the second dielectric material, performing an anisotropic etching process to remove portions of the channel material and portions of the ferroelectric material from the bottoms of the first trenches.

4. The method of claim 3, wherein after performing the anisotropic etching process, the substrate is exposed at the bottoms of the first trenches, wherein the second dielectric material is formed to be in contact with the substrate.

5. The method of claim 2, further comprising, after forming the channel material and before forming the second dielectric material, forming a capping layer over the channel material.

6. The method of claim 5, further comprising, after forming the capping layer and before forming the second dielectric material, forming an oxide liner over the capping layer.

7. The method of claim 6, wherein a first dielectric constant of the capping layer and a second dielectric constant of the channel material are higher than a third dielectric constant of the ferroelectric material.

8. The method of claim 1, wherein the channel material is formed of indium gallium zinc oxide, indium zinc oxide, zinc oxide, indium tin oxide, or indium tungsten oxide.

9. The method of claim 1, further comprising forming an isolation region embedded in the second dielectric material, wherein the isolation region is formed to extend vertically through the layer stack, and to extend laterally between a first sidewall of the layer stack and a second sidewall of the layer stack.

10. The method of claim 9, wherein the isolation region is formed to extend continuously from the first sidewall of the layer stack to the second sidewall of the layer stack.

11. The method of claim 1, wherein forming the SL and the BL comprises:

forming openings in the second dielectric material and extending vertically through the layer stack; and filling the openings with a second electrically conductive material.

12. A method of forming a ferroelectric random access memory (FeRAM) device, the method comprising:

forming a layer stack over a substrate, wherein the layer stack comprises alternating layers of a first dielectric material and an electrically conductive material;

forming first trenches extending vertically through the layer stack;

filling the first trenches with a first structure, wherein the first structure comprises a second dielectric material, a channel material along sidewalls of the second dielectric material, and a ferroelectric material along sidewalls of the channel material;

after filling the first trenches, forming second trenches extending vertically through the layer stack, the second trenches and the first trenches being formed at alternating locations in the layer stack; and

filling the second trenches with a second structure, wherein the second structure has a same layered structure as the first structure.

13. The method of claim 12, further comprising, after filling the second trenches, forming source lines (SLs) and bit lines (BLs) that are embedded at least partially in the second dielectric material, wherein the SLs and the BLs extend vertically through the layer stack.

14. The method of claim 12, further comprising forming isolation regions extending through the layer stack, comprising:

forming openings in the filled first trenches and in the filled second trenches, wherein each of the openings

exposes a first sidewall of a respective trench and a second opposing sidewall of the respective trench; and filling the openings with a third dielectric material.

15. The method of claim 12, wherein the first structure further comprises:

a capping layer between the second dielectric material and the channel material; and

an oxide liner between the second dielectric material and the capping layer.

16. The method of claim 15, wherein the channel material has a higher dielectric constant than the ferroelectric material, wherein the capping layer has a higher dielectric constant than the ferroelectric material.

17. A ferroelectric random access memory device comprising:

a layer stack over a substrate, the layer stack comprising alternating layers of a first dielectric material and an electrically conductive material;

a second dielectric material extending vertically through the layer stack;

a channel layer along sidewalls of the second dielectric material;

a ferroelectric film along sidewalls of the channel layer; and

a source line (SL) and a bit line (BL) that are at least partially embedded in the second dielectric material, wherein the SL and the BL extend through the layer stack.

18. The ferroelectric random access memory device of claim 17, further comprising an isolation region between the SL and the BL, wherein the isolation region extends through the layer stack.

19. The ferroelectric random access memory device of claim 17, further comprising a capping layer between the channel layer and the second dielectric material, wherein a first dielectric constant of the capping layer and a second dielectric constant of the channel layer are higher than a third dielectric constant of the ferroelectric film.

20. The ferroelectric random access memory device of claim 19, wherein a first portion of the ferroelectric film and a first portion of the channel layer are disposed laterally between the capping layer and the layer stack, wherein a second portion of the ferroelectric film and a second portion of the channel layer are disposed vertically between the capping layer and the substrate.

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